Data Science with Python

Unsupervised Learning



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10 Common ways to Measure Association

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What is Unsupervised Learning?

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What is Unsupervised Learning?

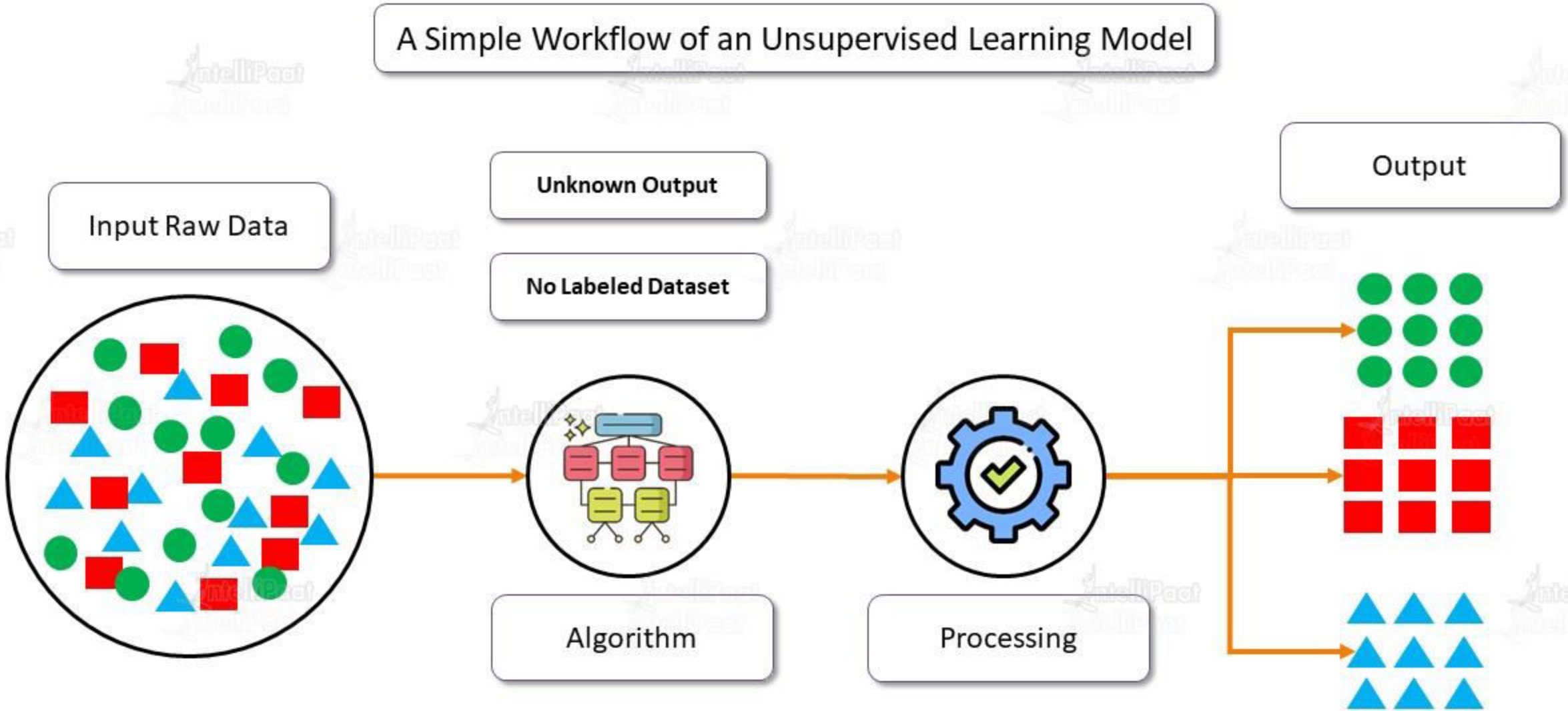
Using unsupervised learning, we can look for the previously undetected trends in a given dataset without any labeled data

It is called '**Unsupervised**' because it needs little or no human supervision over the learning process



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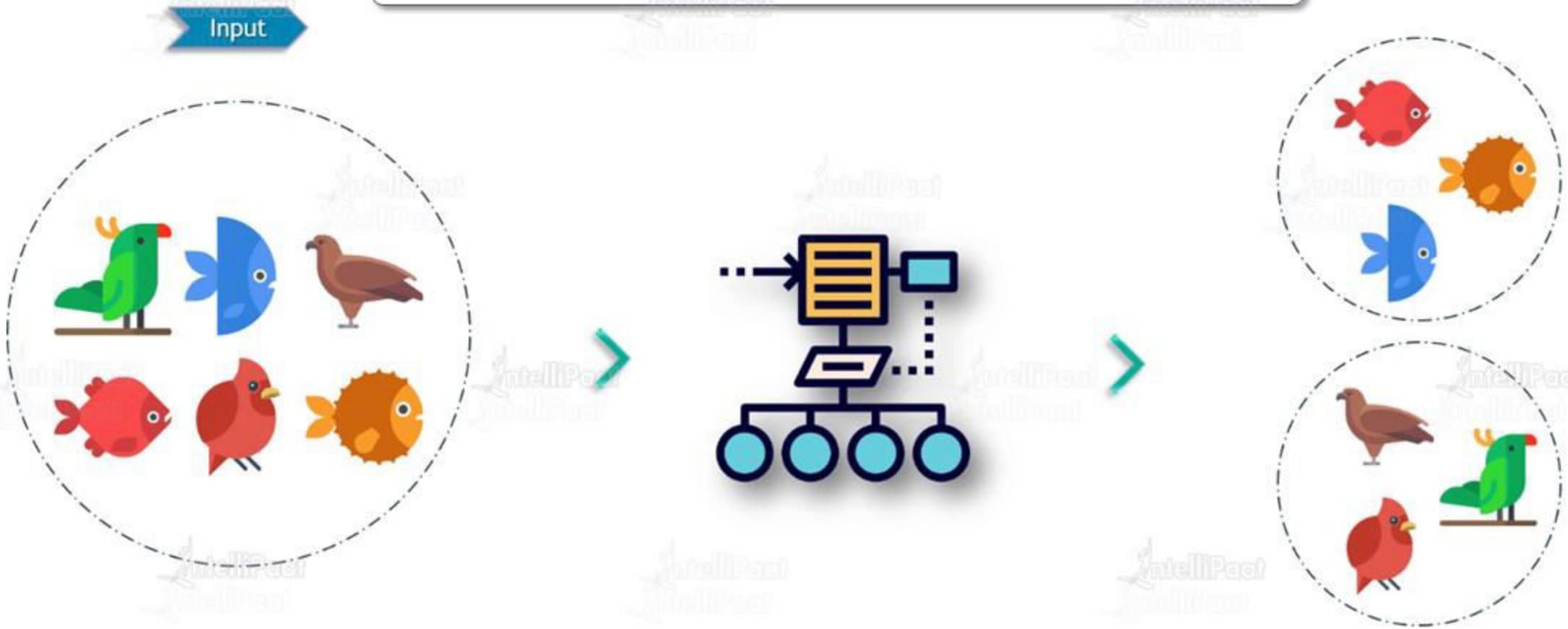
What is Unsupervised Learning?



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What is Unsupervised Learning?

A Simple Workflow of an Unsupervised Learning Model



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What is Clustering?

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What is Clustering?

The process of dividing datasets into groups of similar data points



Clustering is dividing a dataset into data points where:

- Points in the same group are as similar as possible
- Points in the different groups are as dissimilar as possible

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Use Cases of Clustering

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Use Cases of Clustering

A Few Clustering Use Cases

Day-to-day grouping activities that can speed up our shopping

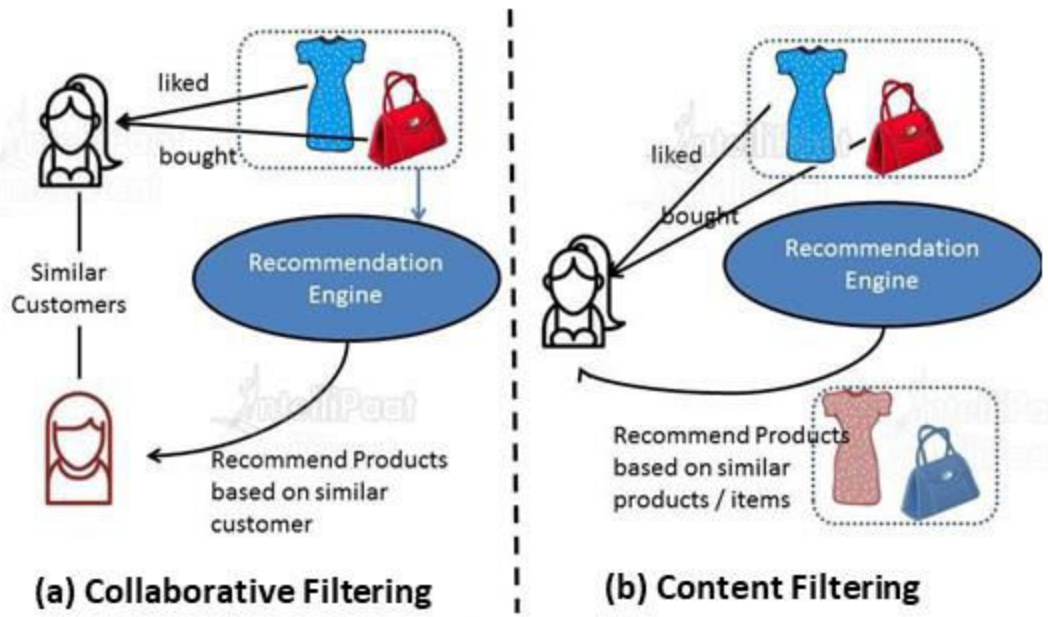


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Use Cases of Clustering

A Few Clustering Use Cases

Recommending products based on our search history



amazon

Flipkart

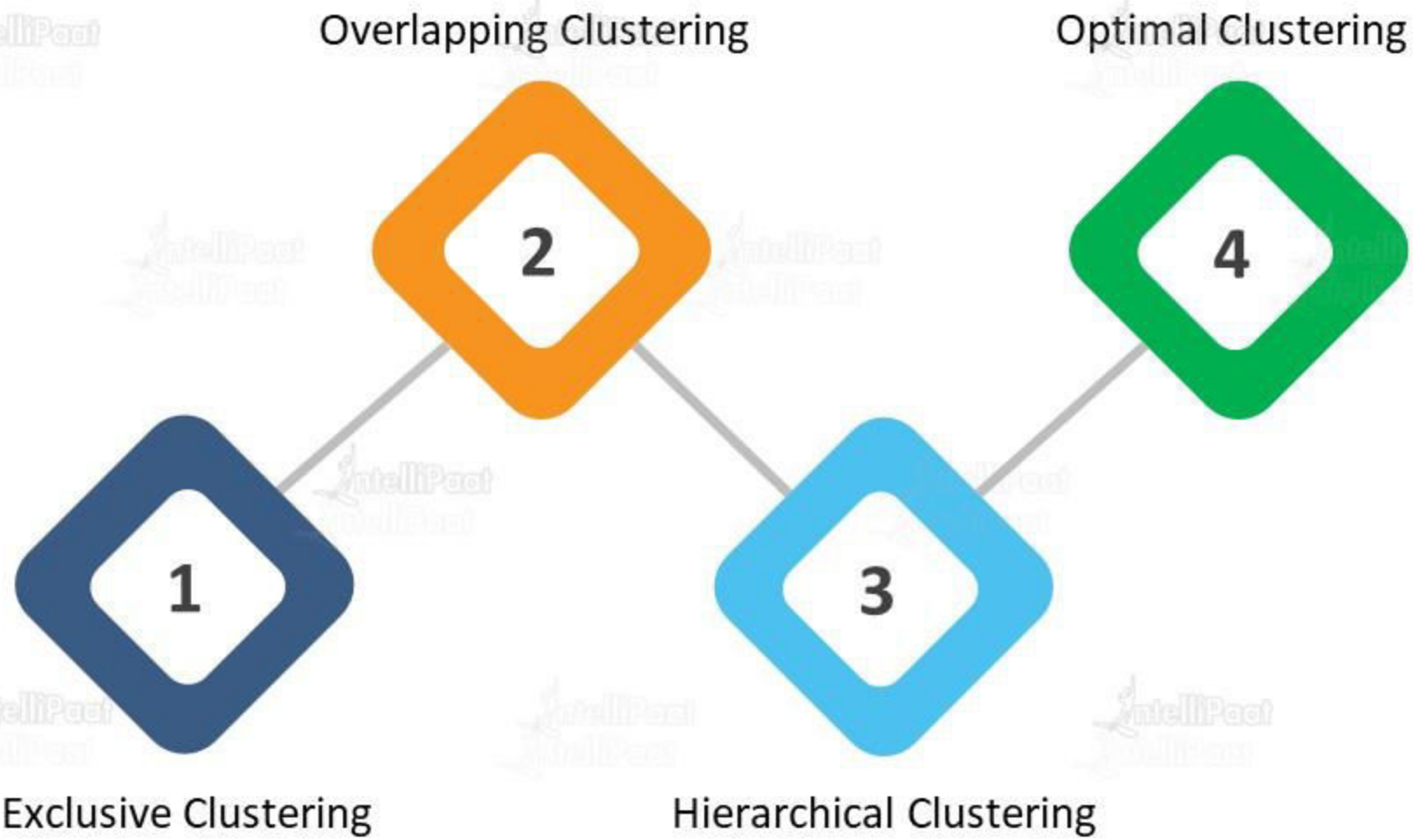
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Types of Clustering

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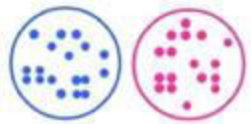
Types of Clustering



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Types of Clustering

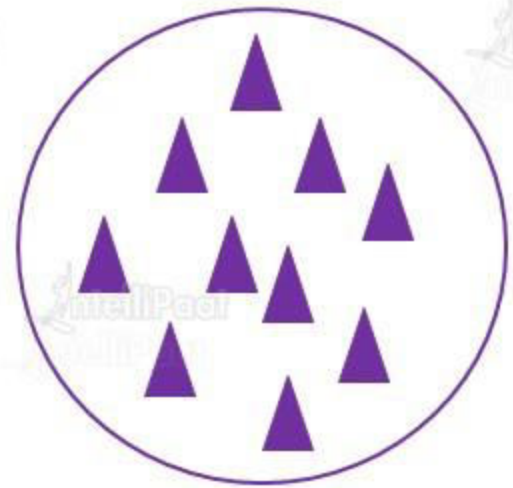
- Exclusive Clustering
- Overlapping Clustering
- Hierarchical Clustering
- Optimal Clustering



Each data object can only exist in a single cluster. It is are grouped in a certain way that the data belongs to only one definite cluster

For example: K-means clustering

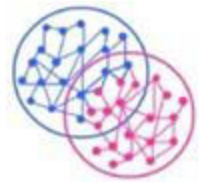




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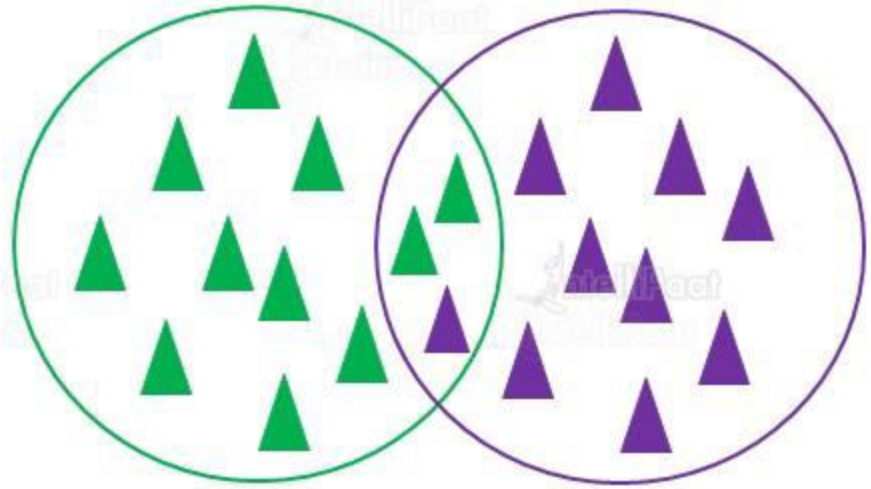
Types of Clustering

- Exclusive Clustering
- Overlapping Clustering
- Hierarchical Clustering
- Optimal Clustering



It allows data objects to be grouped in 2 or more clusters. In fuzzy clustering, e.g., every data object belongs to every cluster

For example: Fuzzy/C-means clustering

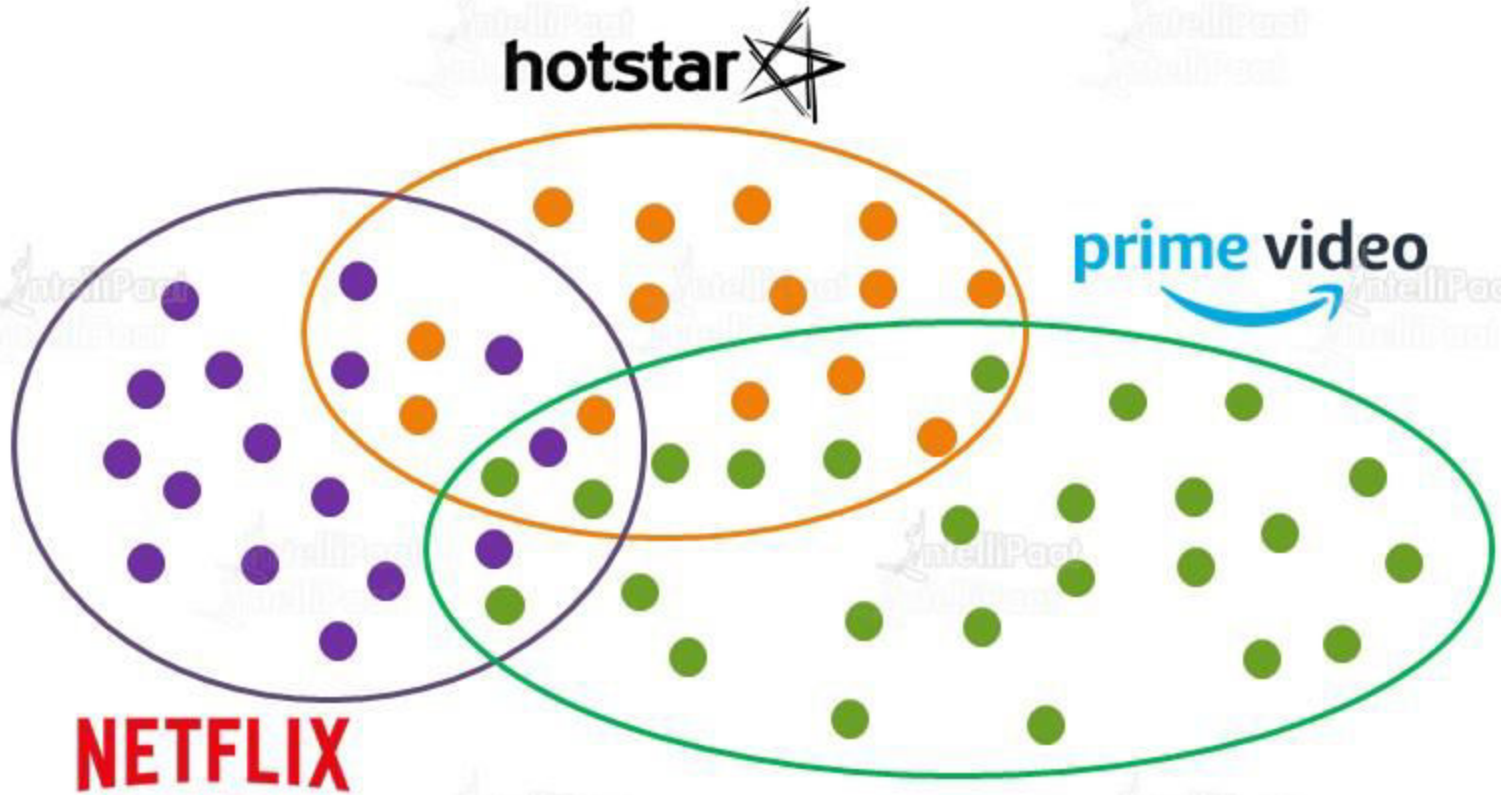


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Types of Clustering

- Exclusive Clustering
- Overlapping Clustering
- Hierarchical Clustering
- Optimal Clustering

Suppose, a company wants to market a streaming platform differently to Hotstar, Netflix, and Prime users. There will also be users who use two or all three of these services

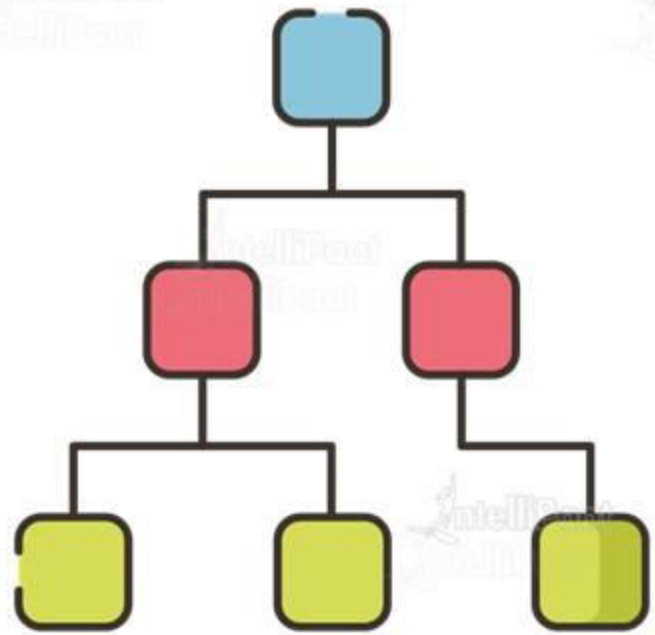


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Types of Clustering

- Exclusive Clustering
- Overlapping Clustering
- Hierarchical Clustering
- Optimal Clustering

Hierarchical clustering is a cluster analysis used to build a hierarchy of various clusters. The clusters will be in a predominant order from top to bottom

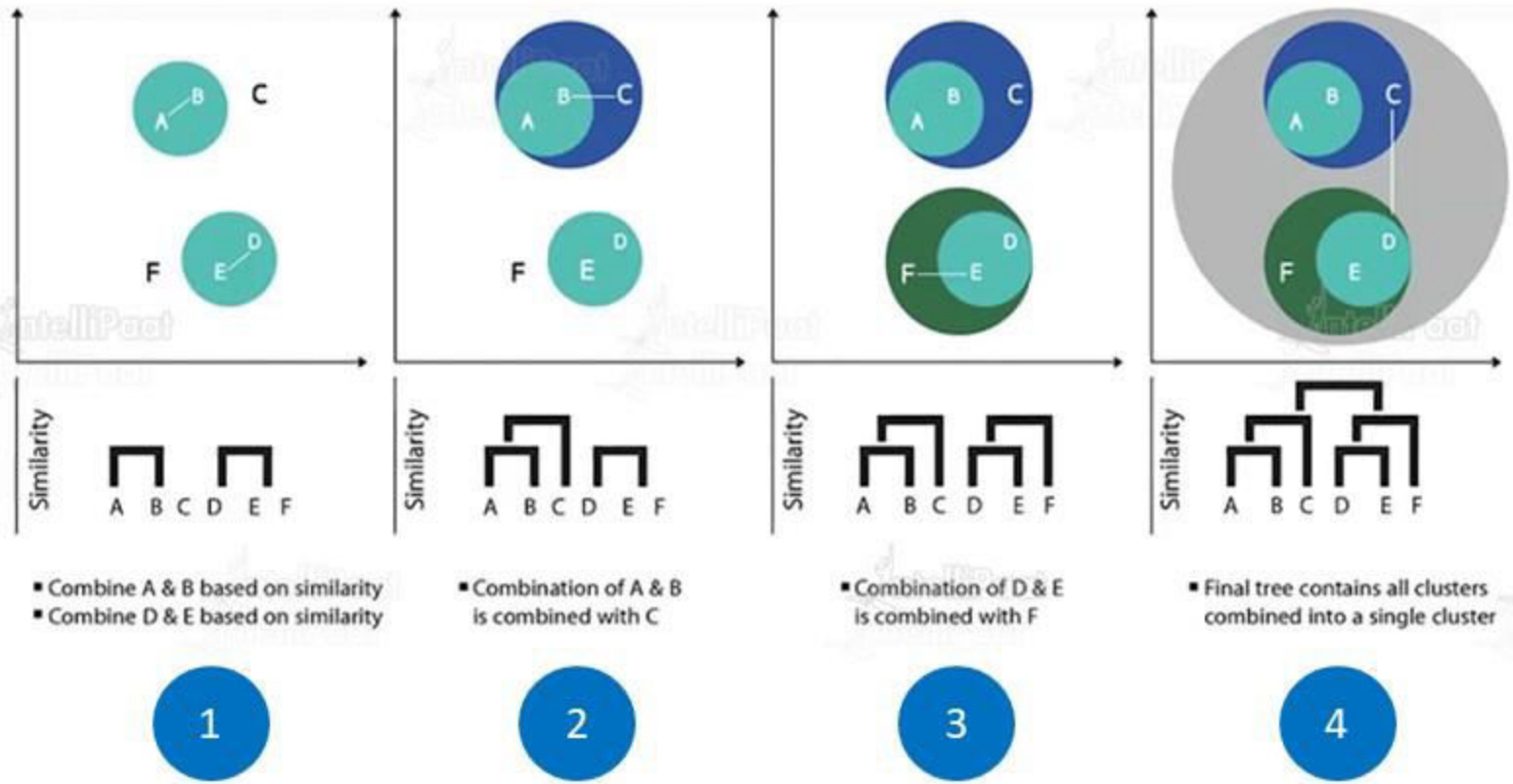


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Types of Clustering

- Exclusive Clustering
- Overlapping Clustering
- Hierarchical Clustering
- Optimal Clustering

Steps to create a hierarchical cluster:

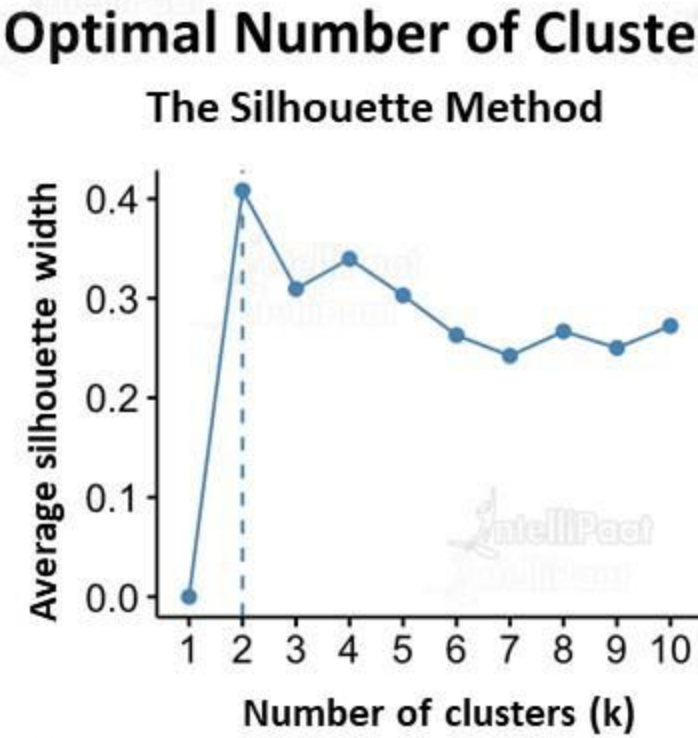


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Types of Clustering

- Exclusive Clustering
- Overlapping Clustering
- Hierarchical Clustering
- Optimal Clustering

Optimal clustering measures the quality of clustering. It is used to find the optimal number (k) of clusters



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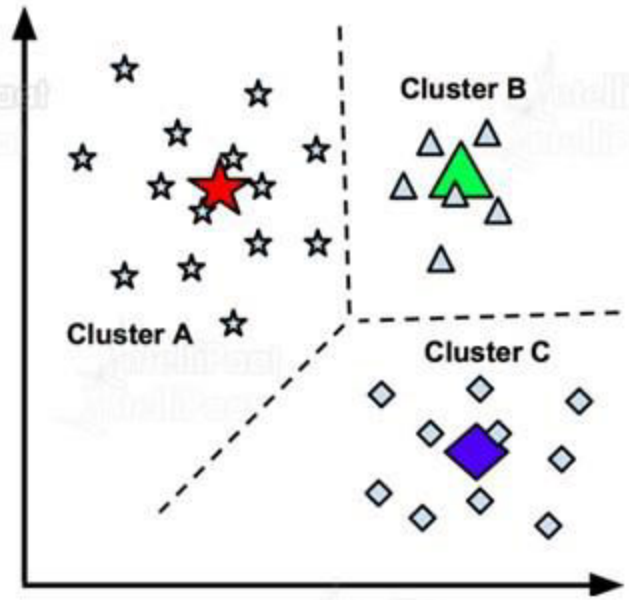
What is K-means Clustering?

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What is K-means Clustering?

K-means is a clustering algorithm that focuses on grouping similar elements or data points into a cluster

'**K**' in k-means represents the number of clusters



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What is K-means Clustering?

Business Applications of K-means Clustering

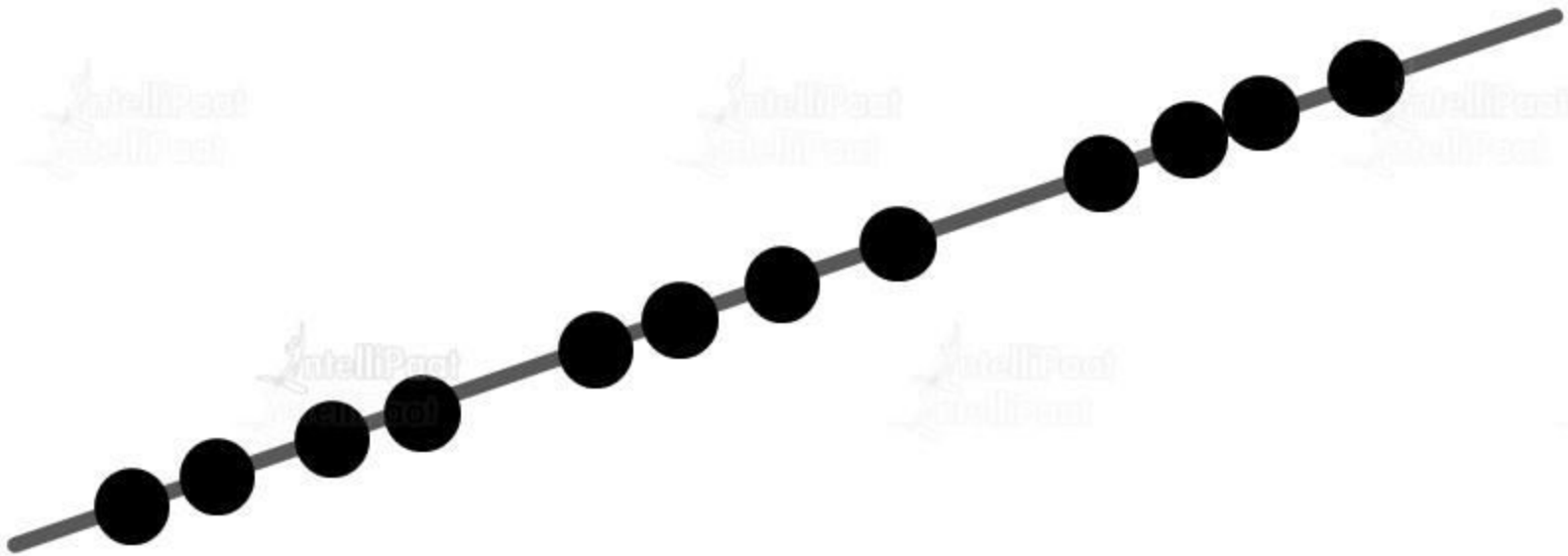
- Behavioral segmentation
- Inventory categorization
- Sorting sensor measurements
- Detecting bots or anomalies

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K-means Clustering Example

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K-means Clustering Example

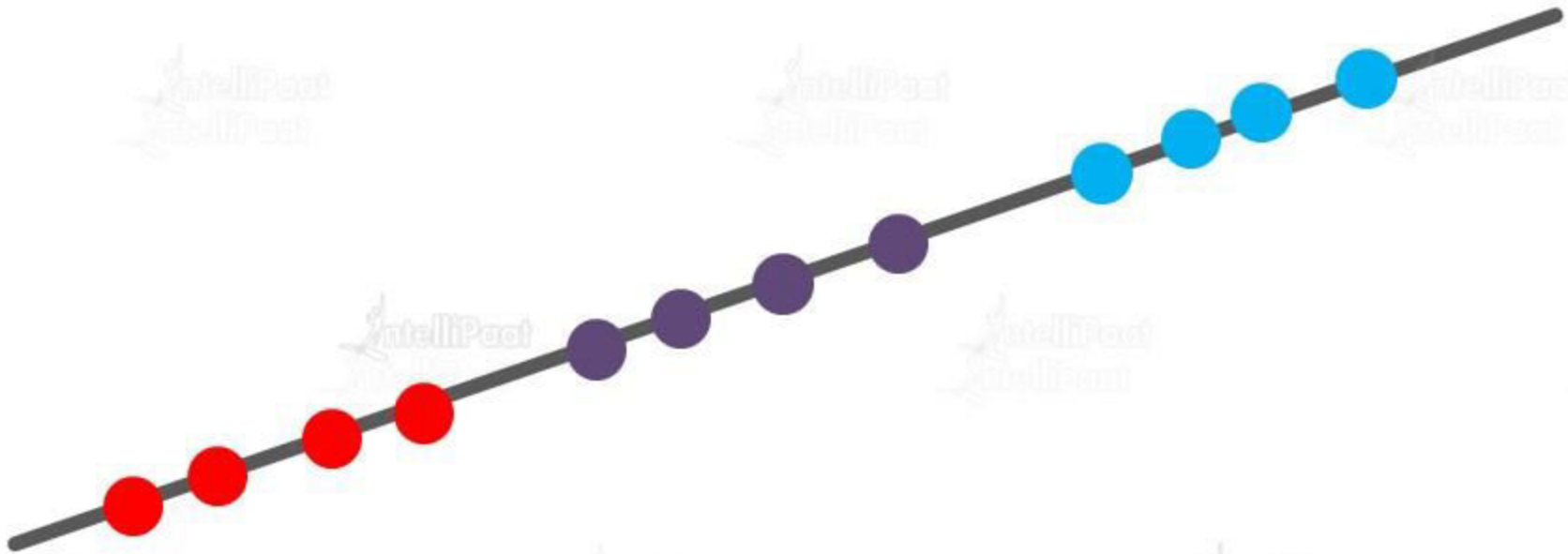


Number of Clusters = 3

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K-means Clustering Example

This is our final solution out of this K-means clustering example



Number of Clusters, k = 3

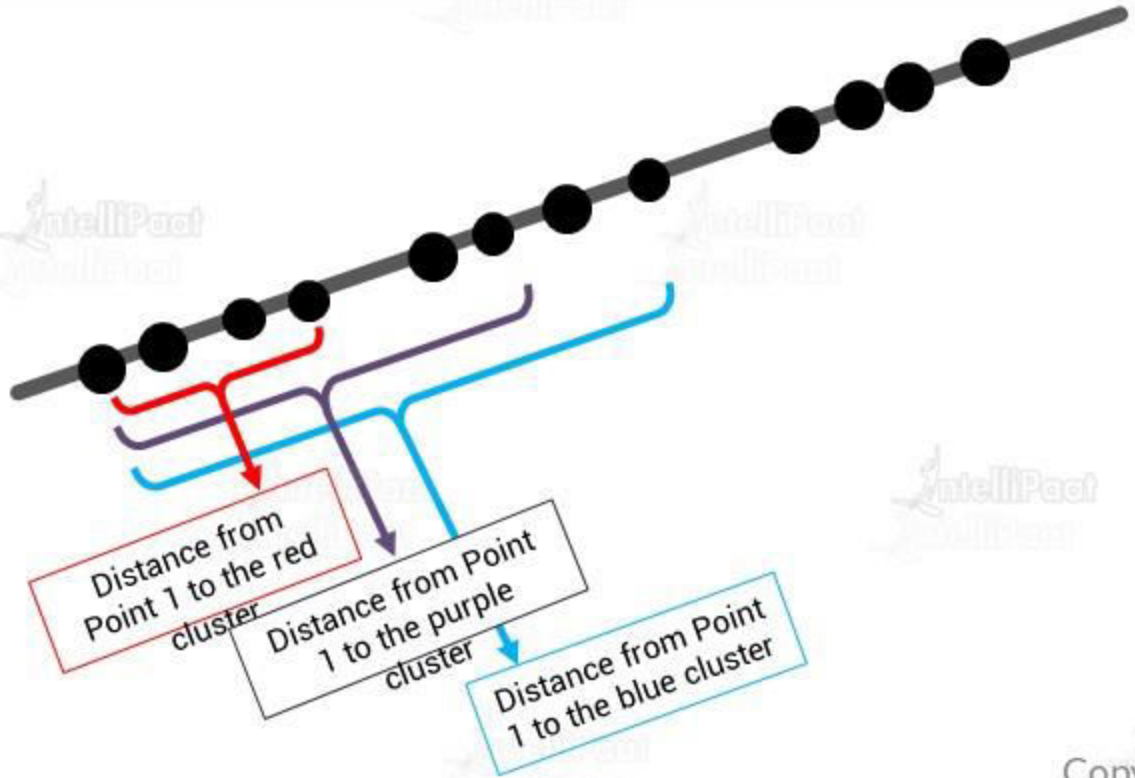
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K-means Clustering Example

Step 1: Select the number of clusters to be identified, i.e., select a value for k (in this case, **k = 3**)

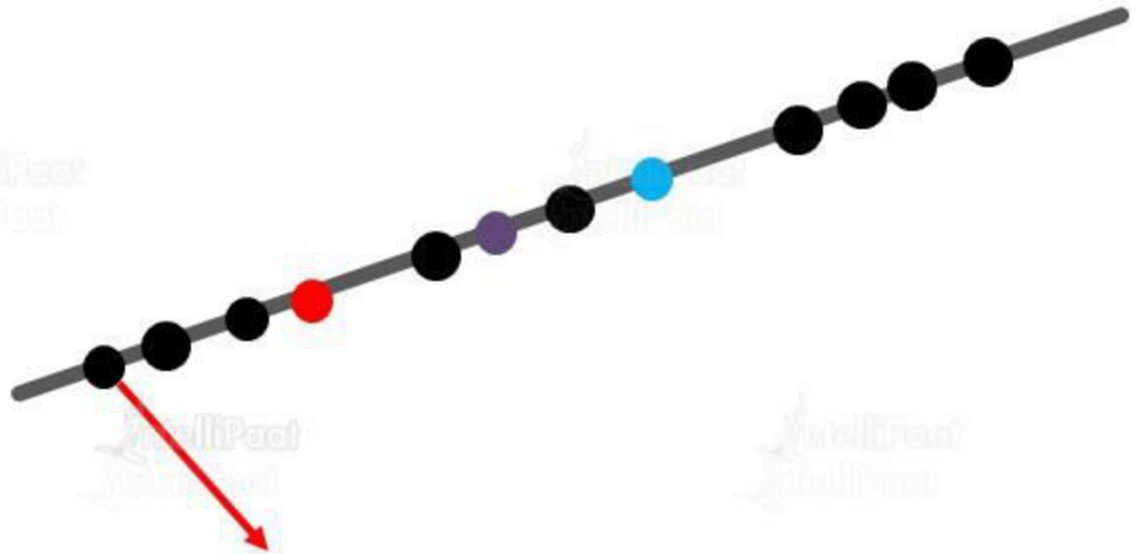
Step 2: Randomly, select three distinct data points

Step 3: Measure the distance between the first point and the select three clusters



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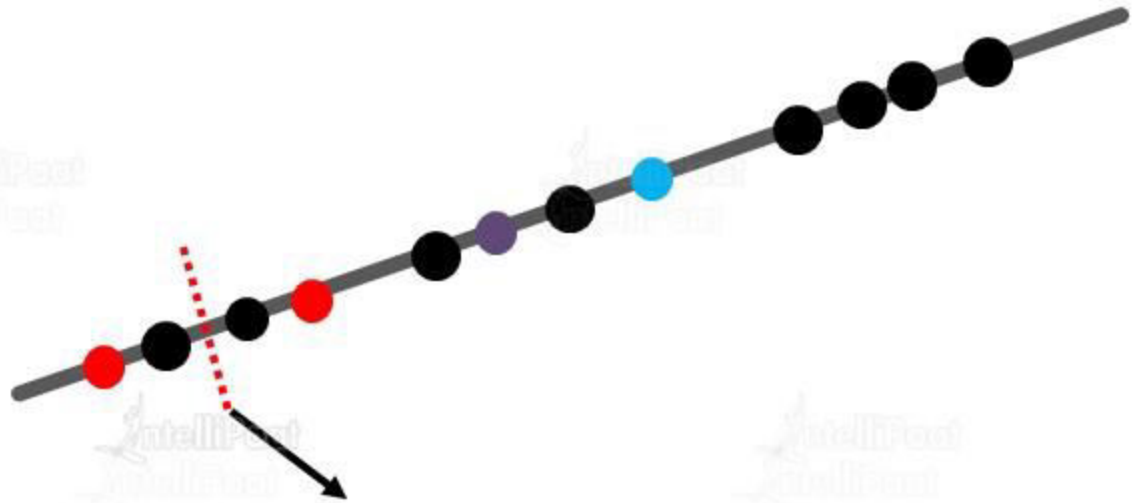
K-means Clustering Example



Step 4: Assign the first point to the nearest cluster (red in our case)

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K-means Clustering Example

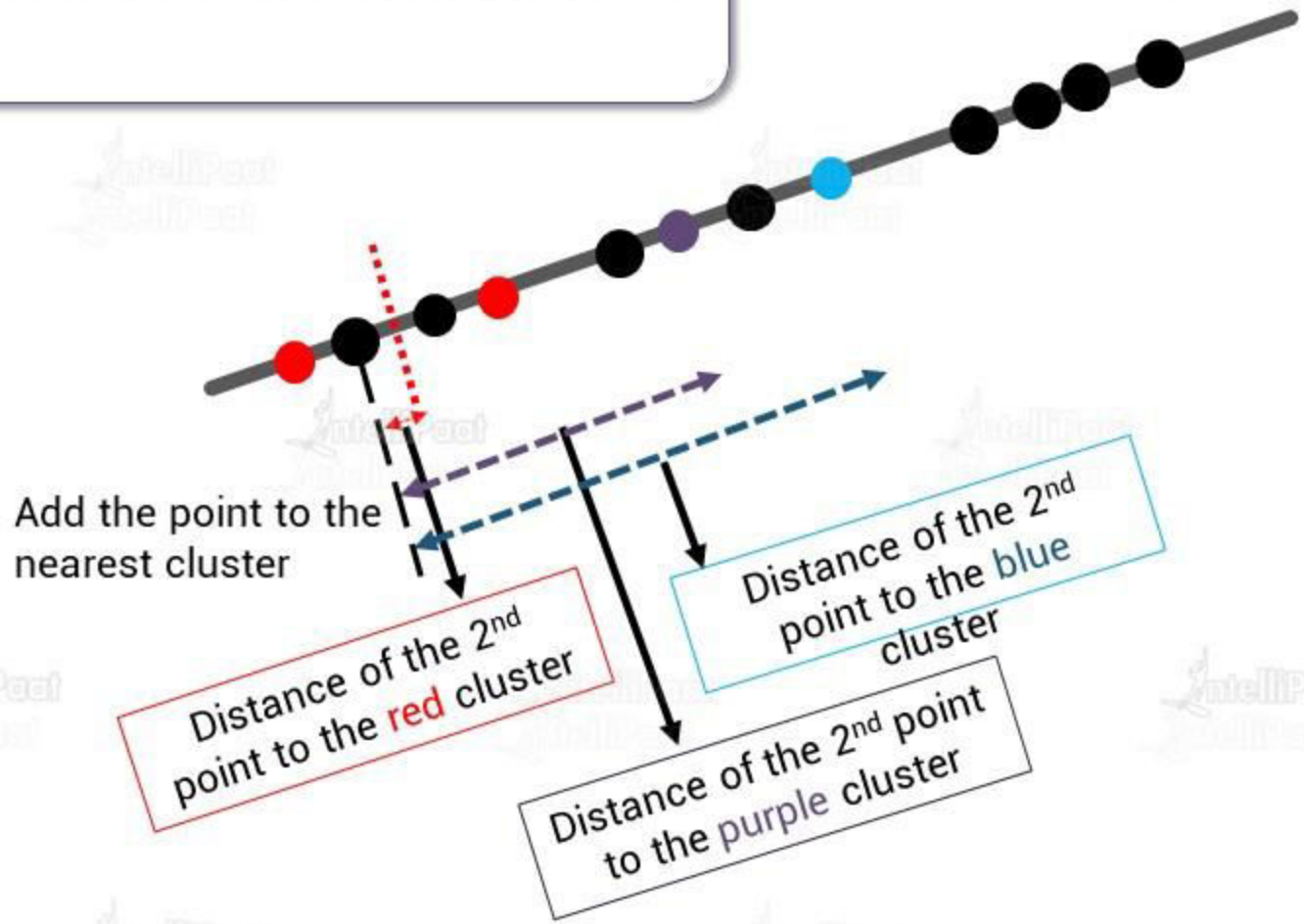


Step 5: Calculate the mean value, including the new point, for the red cluster

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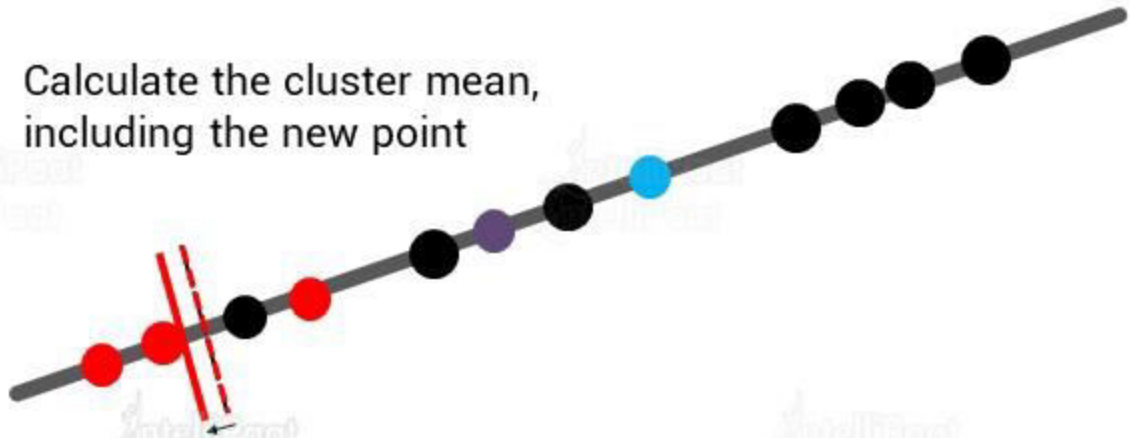
K-means Clustering Example

Find which cluster Point 2 belongs to and how and then repeat the same procedure but measuring the distance to the red mean



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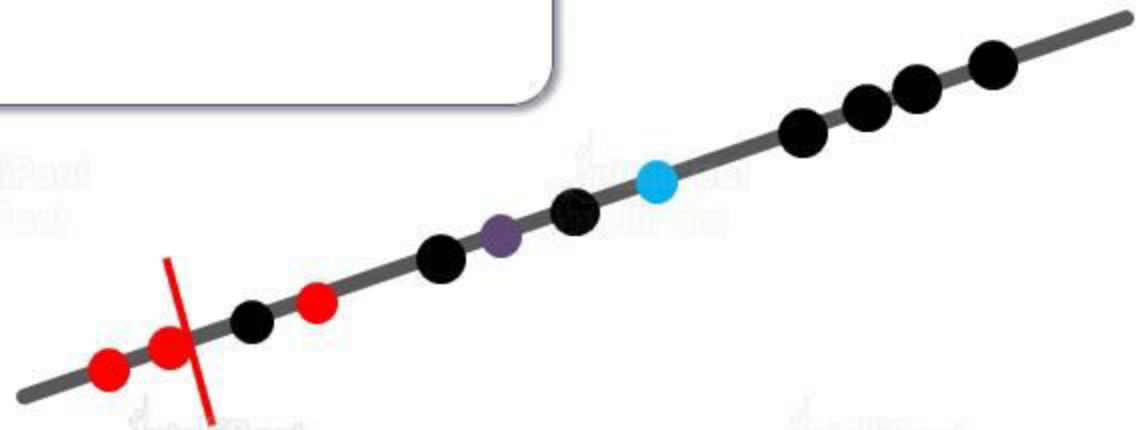
K-means Clustering Example



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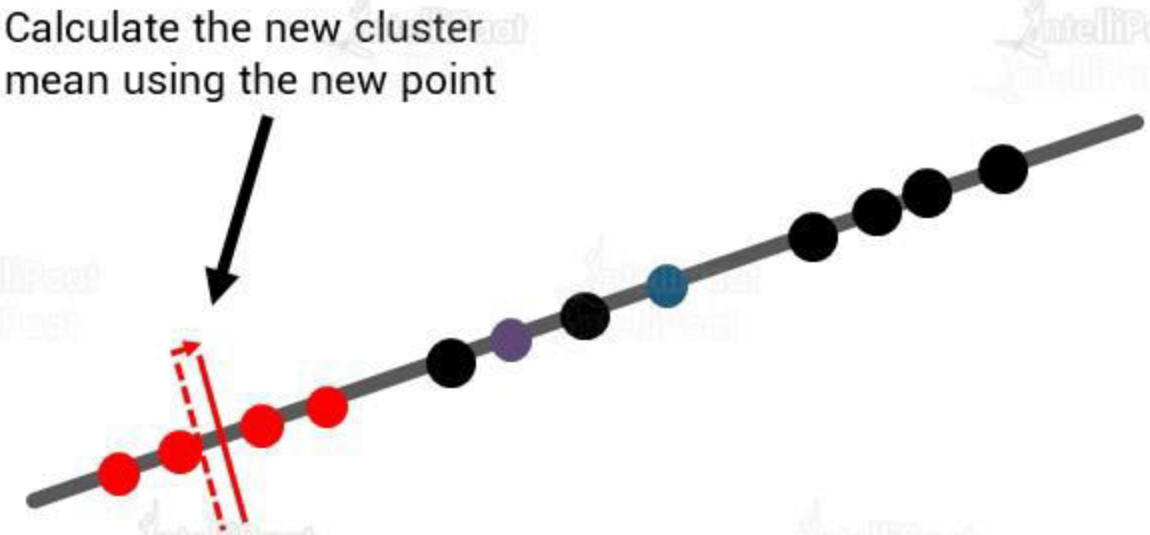
K-means Clustering Example

Find which cluster Point 3 belongs to and how and then repeat the same procedure but measure the distance from the new red mean



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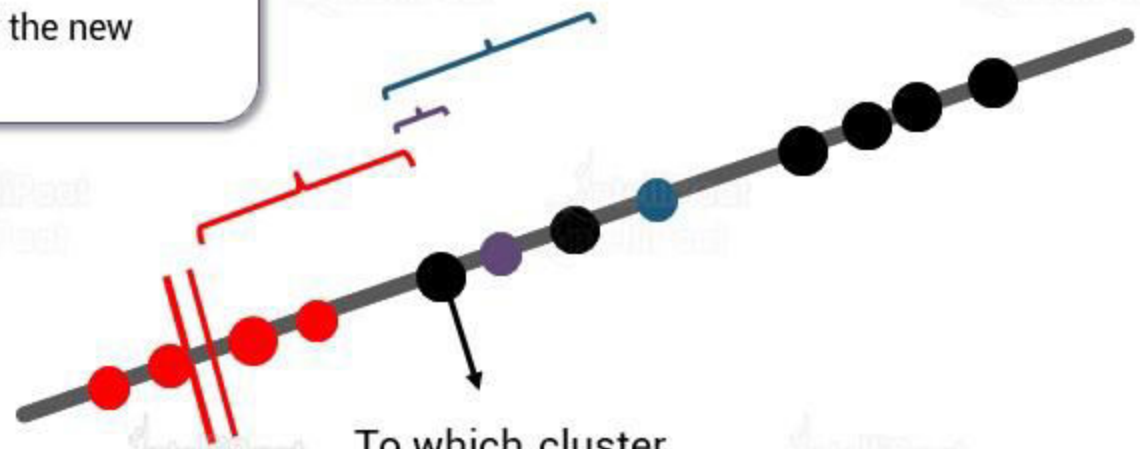
K-means Clustering Example



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K-means Clustering Example

1. **Measure the distance**
2. Assign the point to the nearest cluster
3. Calculate the cluster mean using the new point



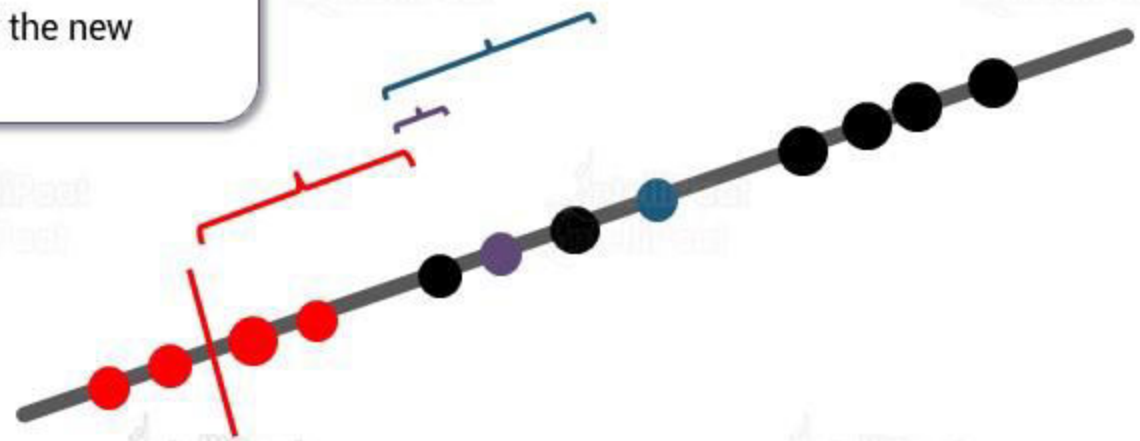
To which cluster does this point belong?

Repeat the steps again

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K-means Clustering Example

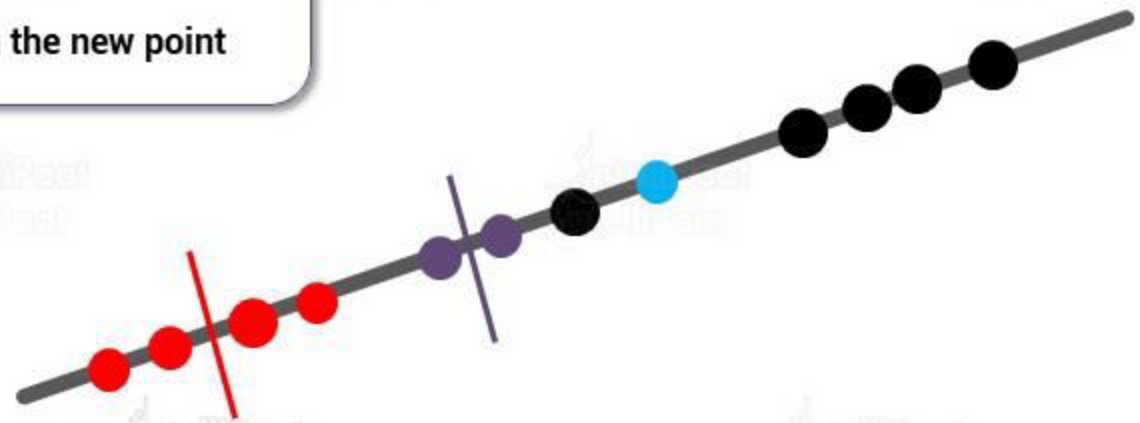
1. Measure the distance
2. **Assign the point to the nearest cluster**
3. Calculate the cluster mean using the new point



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K-means Clustering Example

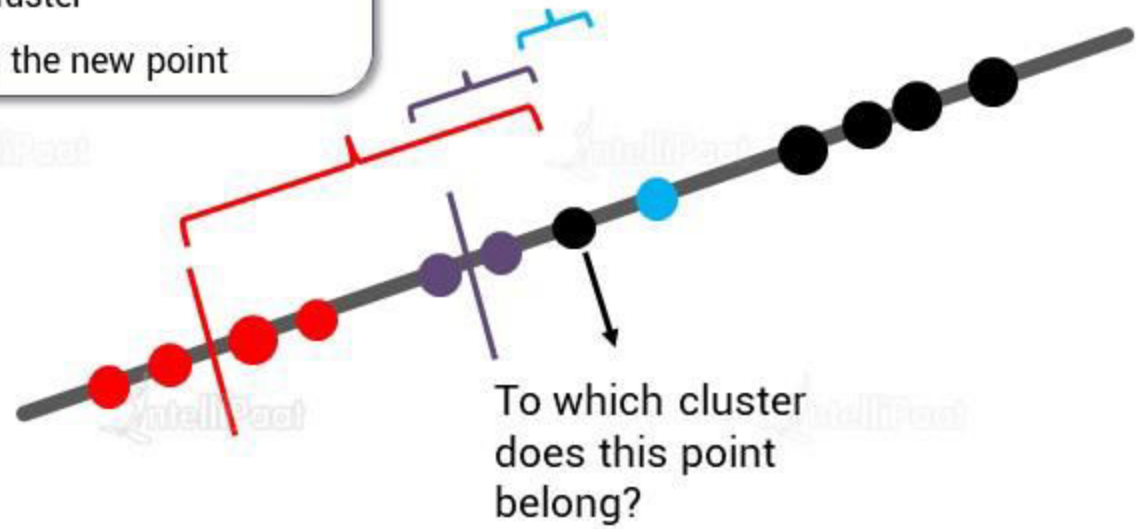
1. Measure the distance
2. Assign the point to the nearest cluster
3. **Calculate the cluster mean using the new point**



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K-means Clustering Example

1. **Measure the distance from the cluster mean (centroids)**
2. Assign the point to the nearest cluster
3. Calculate the cluster mean using the new point

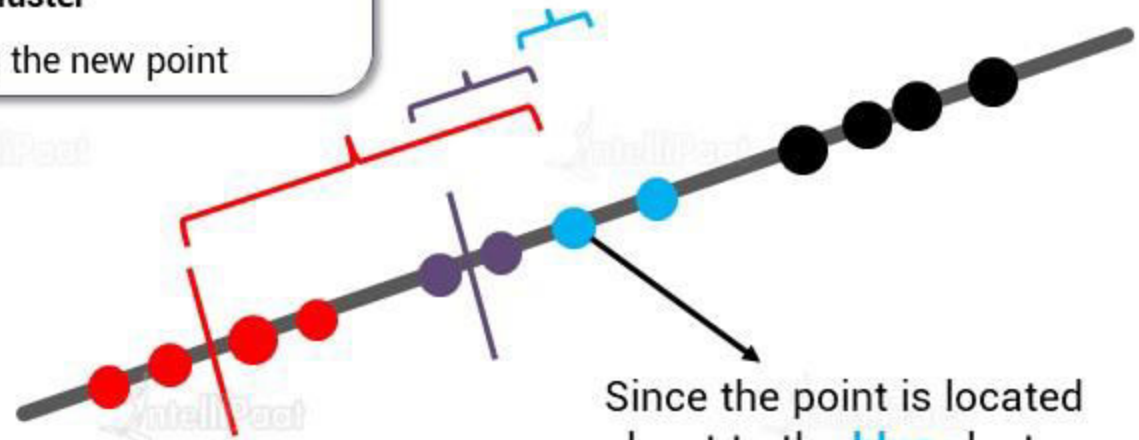


Repeat the same steps until all clusters are assigned

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K-means Clustering Example

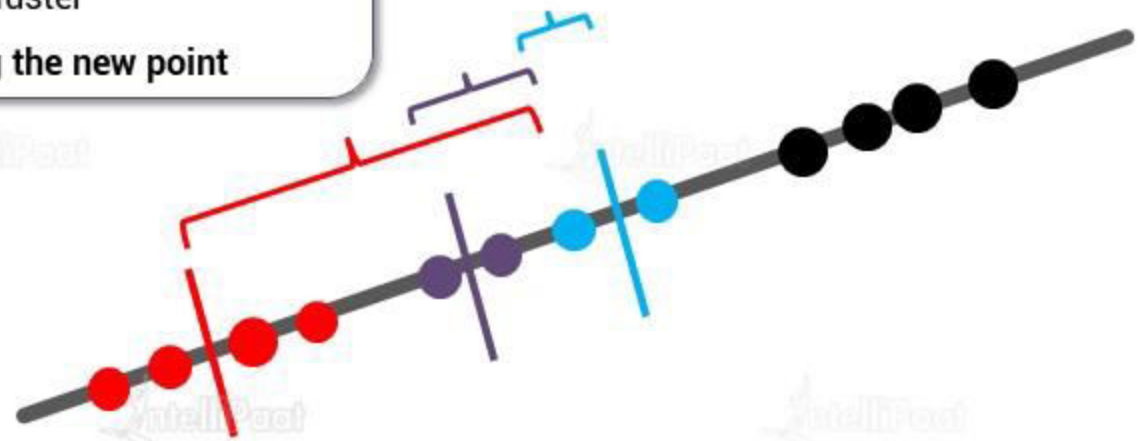
1. Measure the distance from the cluster mean (centroids)
2. **Assign the point to the nearest cluster**
3. Calculate the cluster mean using the new point



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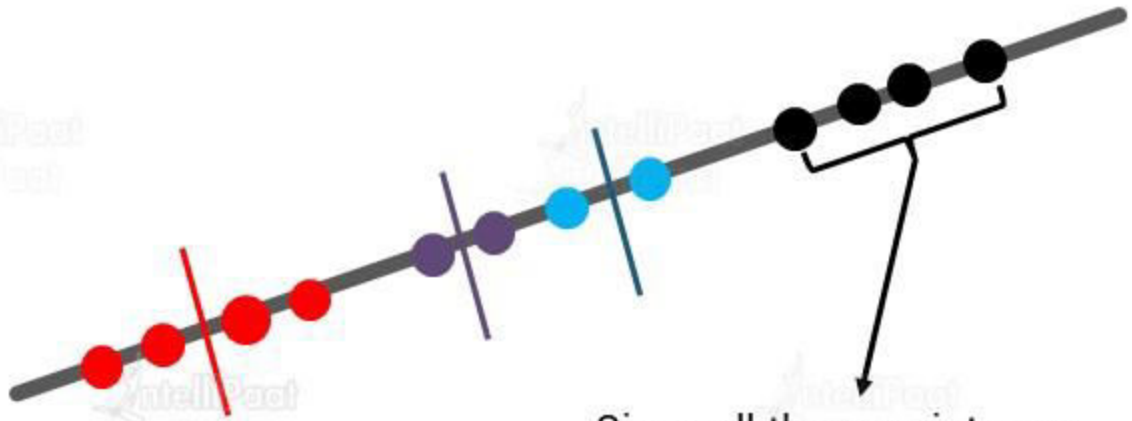
K-means Clustering Example

1. Measure the distance from the cluster mean (centroids)
2. Assign the point to the nearest cluster
3. **Calculate the cluster mean using the new point**



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K-means Clustering Example



Since all these points are located closet to the **blue** cluster, all of them will be assigned to it

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K-means Clustering Example



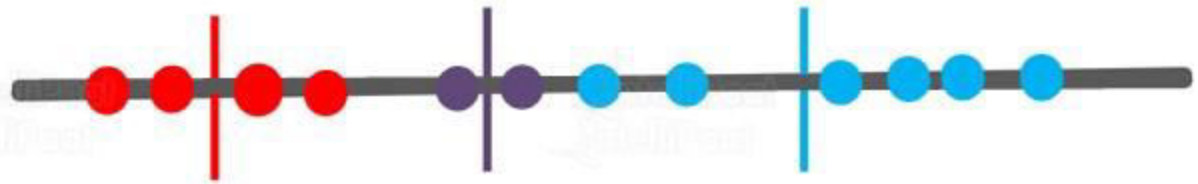
Result from the First Iteration



Original/Expected Result

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K-means Clustering Example



Total Variation Within the Cluster

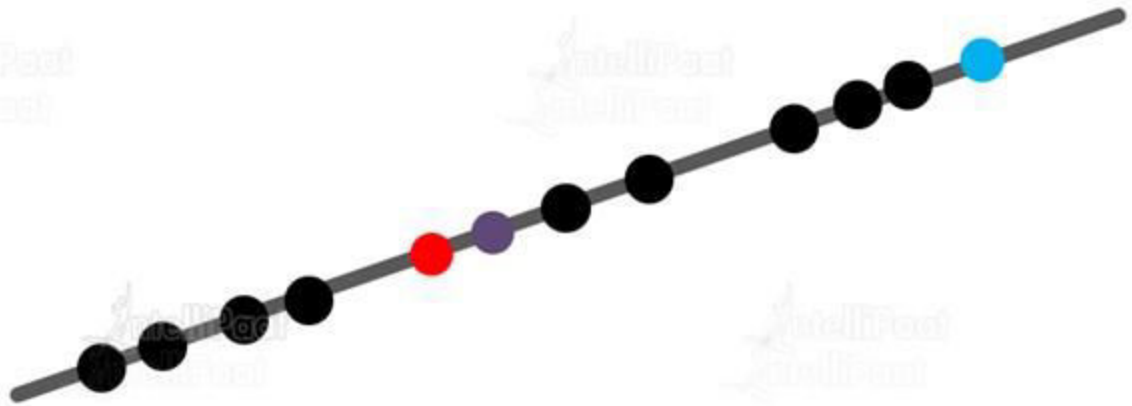
According to the k-means clustering, the algorithm iterates over and over again until the data points within each cluster stop changing

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K-means Clustering Example

Iteration 2:

Again, we will start from scratch. But this time, we will select different initial random points (as compared to the first iteration



Step 1: Select the number of clusters to be identified, i.e., k =3

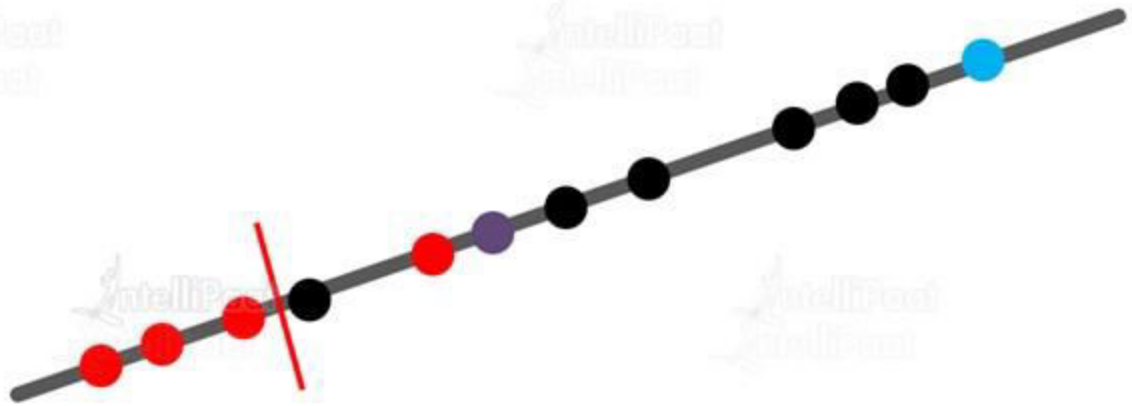
Step 2: Randomly select three distinct data points

Step 3: Measure the distance between the first point and the select three clusters

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K-means Clustering Example

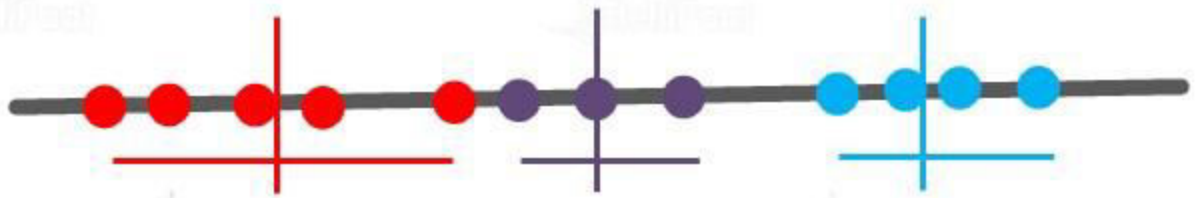
Algorithm picks three initial clusters and adds the remaining points to the cluster with the nearest mean, and recalculating the mean, each time a new point is added to the cluster



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K-means Clustering Example

Algorithm picks three initial clusters and adds the remaining points to the cluster with the nearest mean, and recalculating the mean, each time a new point is added to the cluster



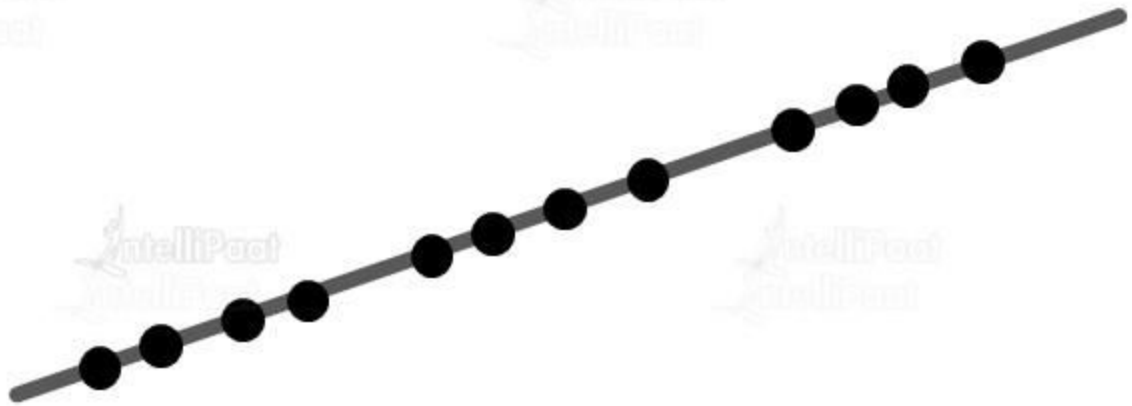
Total Variation Within the Cluster

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K-means Clustering Example

Iteration 3:

Again, we will start from scratch and select different initial random points (as compared to the first and second iterations)



Pick three initial clusters

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K-means Clustering Example

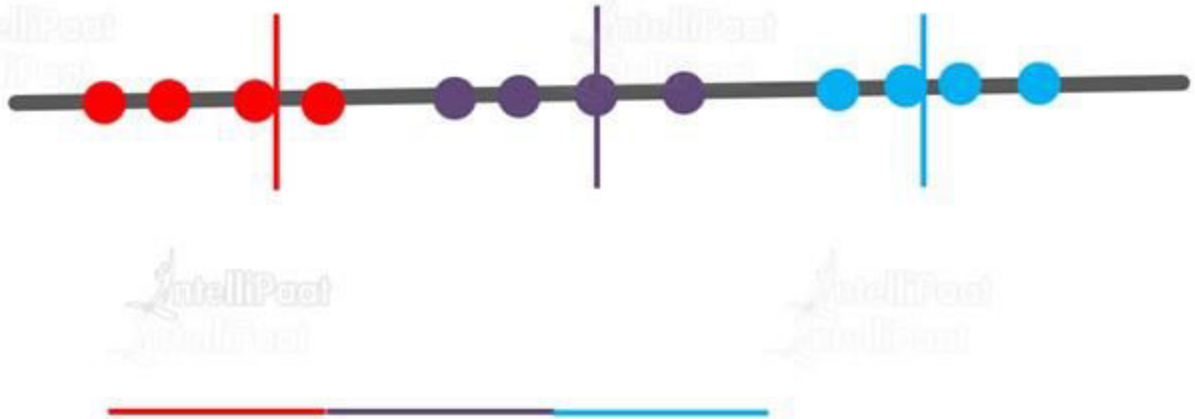


Cluster the remaining points

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K-means Clustering Example

Finally, sum the variation within each other

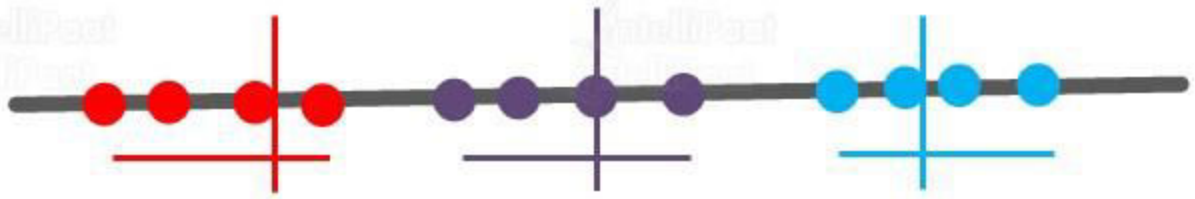


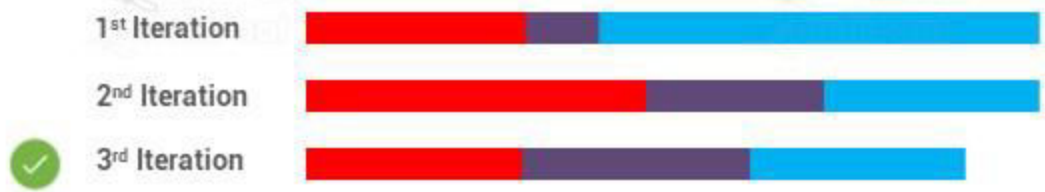
The Total Variation Within the Cluster

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K-means Clustering Example

Finally, sum the variation within each other





But, how to find the value of 'k'?

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K-means Clustering Example

How to find the value of 'k'?

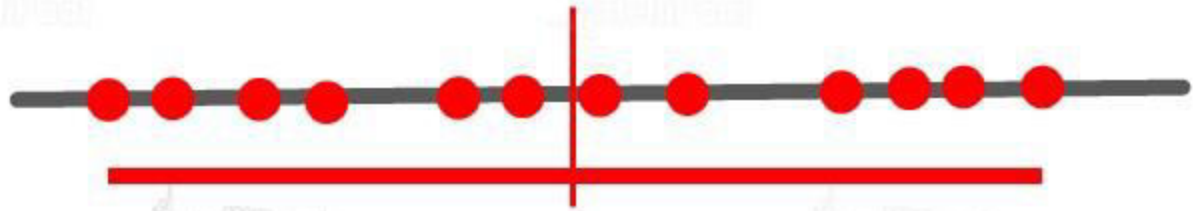
In the previous scenario, we knew that k = 3, but what if we don't know the exact value of k?



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K-means Clustering Example

For finding the value of k, we will use the hit and trial method, **starting from k = 1**

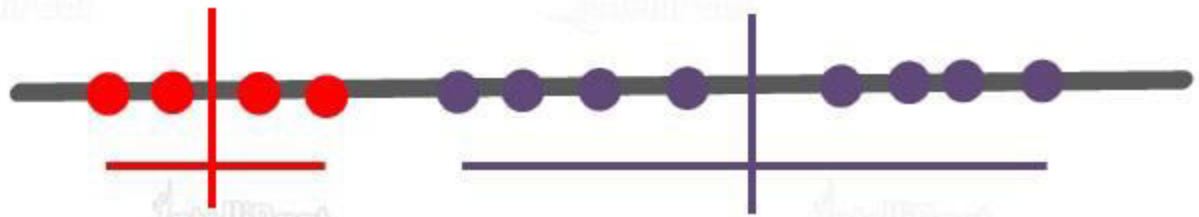


K = 1 is the worst case scenario, even when we cross-verify it with its total variation (all red)

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K-means Clustering Example

So, now, we will try with **k = 2**



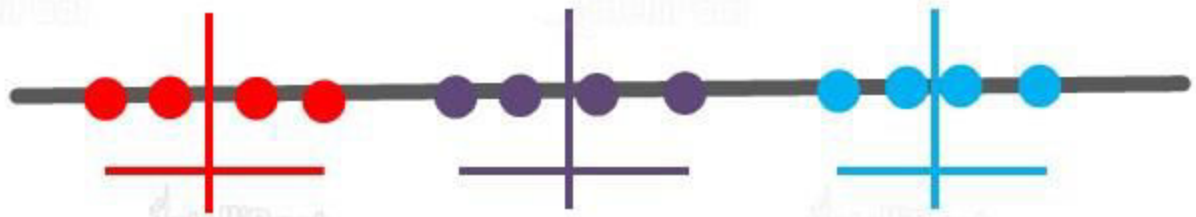
We can see that k = 2 is better than k = 1 (Total Variation)



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K-means Clustering Example

Now, when we try with **k = 3**



It is clear that k = 3 is still better than k = 2 (Total Variation)

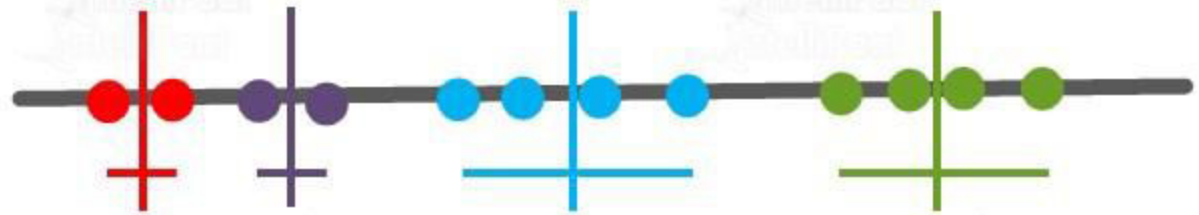


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K-means Clustering Example

Now, try with **k = 4**

- Every time we increase the number of clusters, the variation decreases
- If the number of clusters = the number of data points, variation = 0



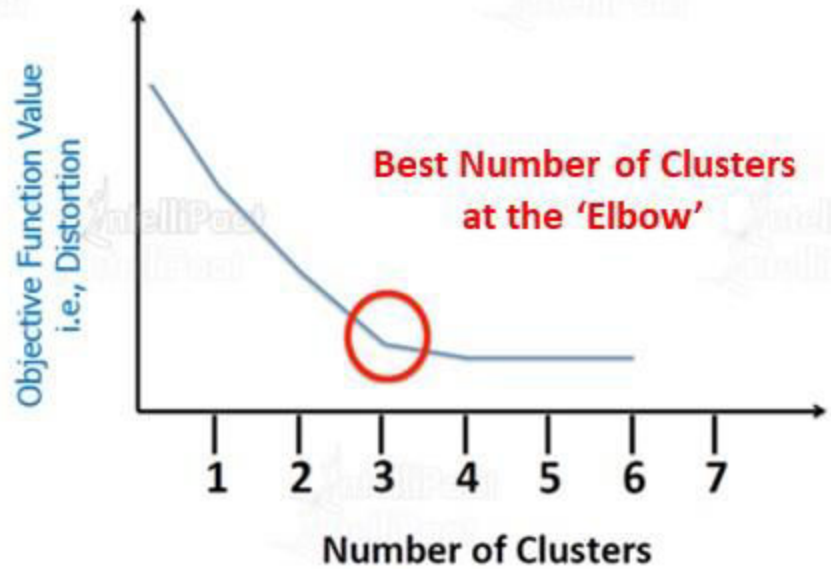
K = 4 is still better than k = 3 (Total Variation)



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K-means Clustering Example

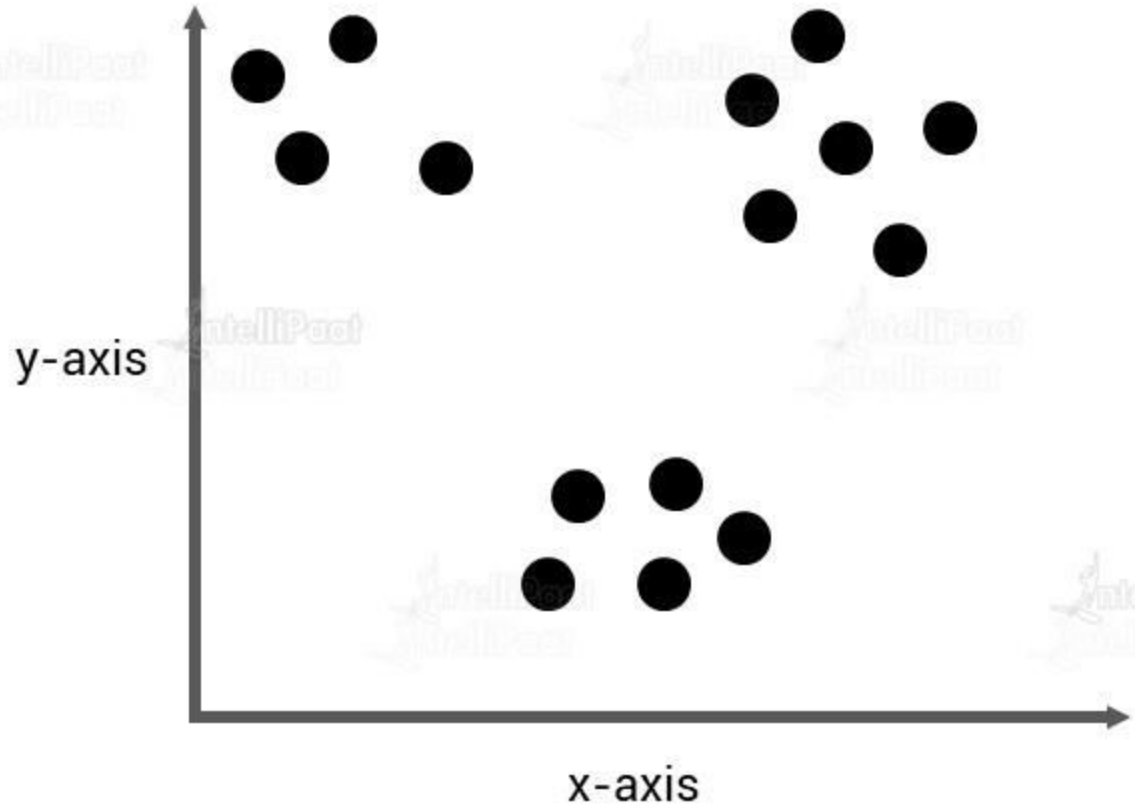
Finally, we got the best number of clusters at the elbow, which is **3**



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K-means Clustering Example

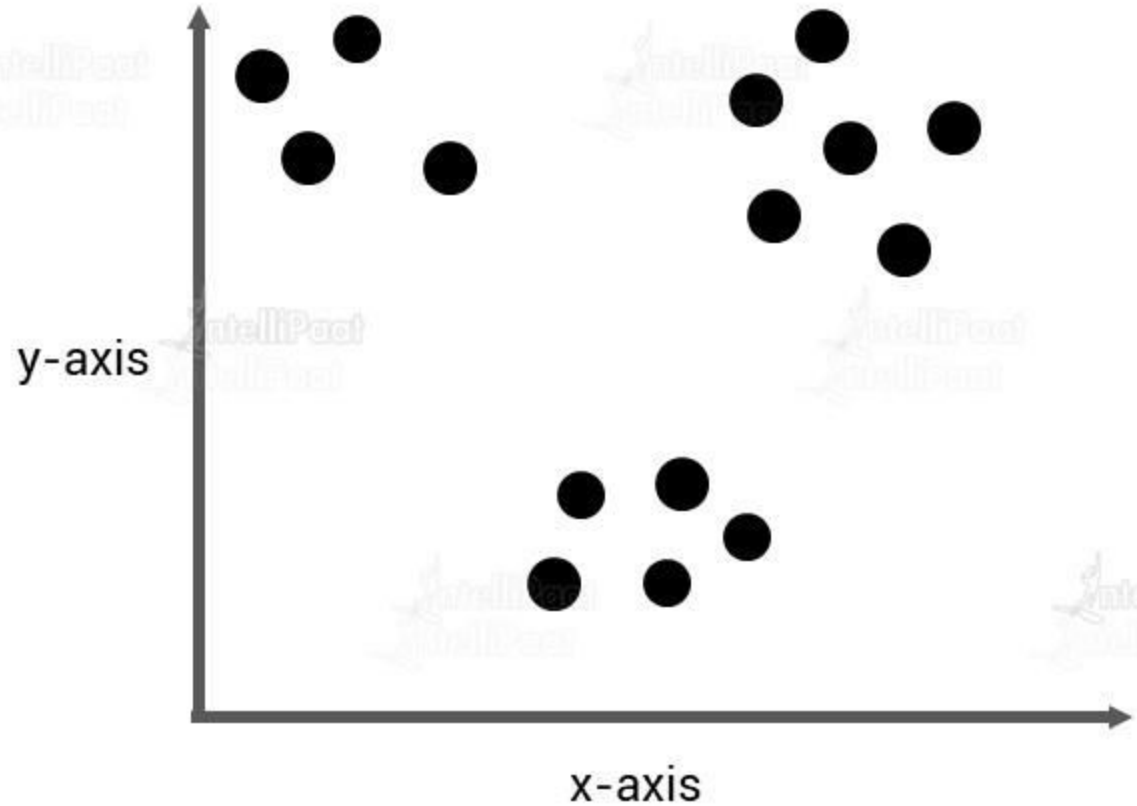
Now, what if we have our data plotted on the x and y axes



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K-means Clustering Example

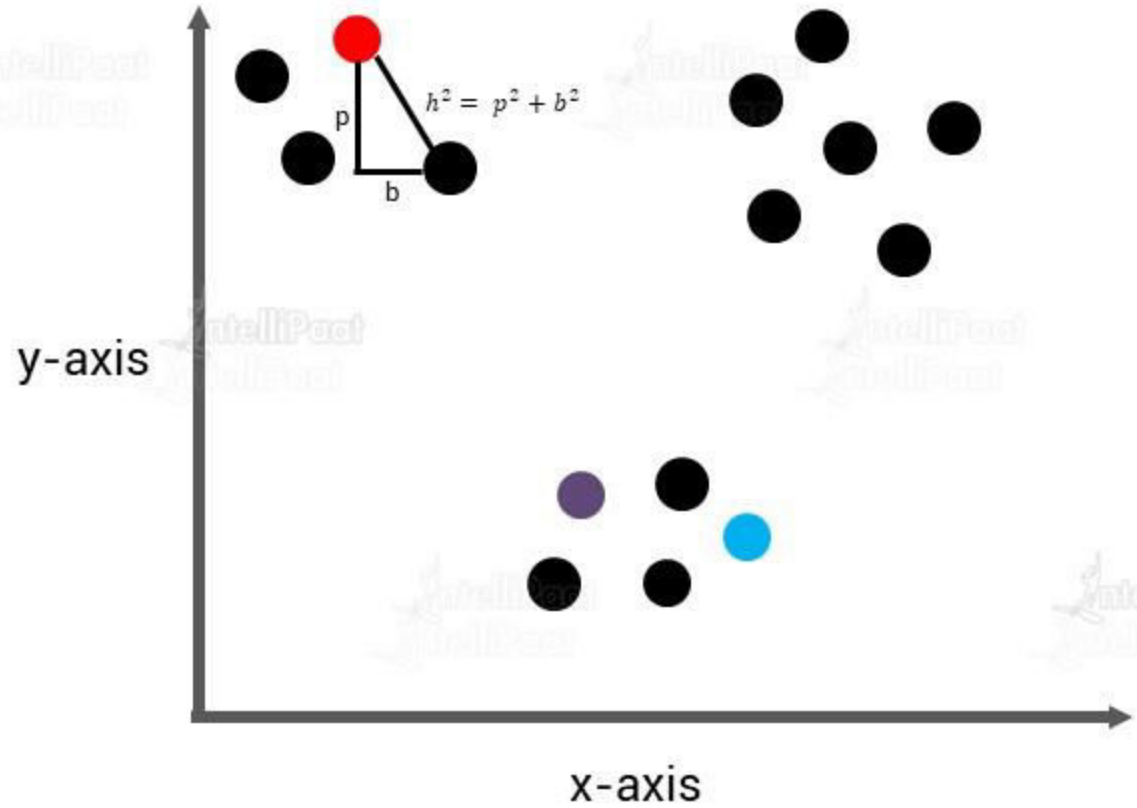
In the same way as mentioned, we have to pick three random points in any of the clusters



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K-means Clustering Example

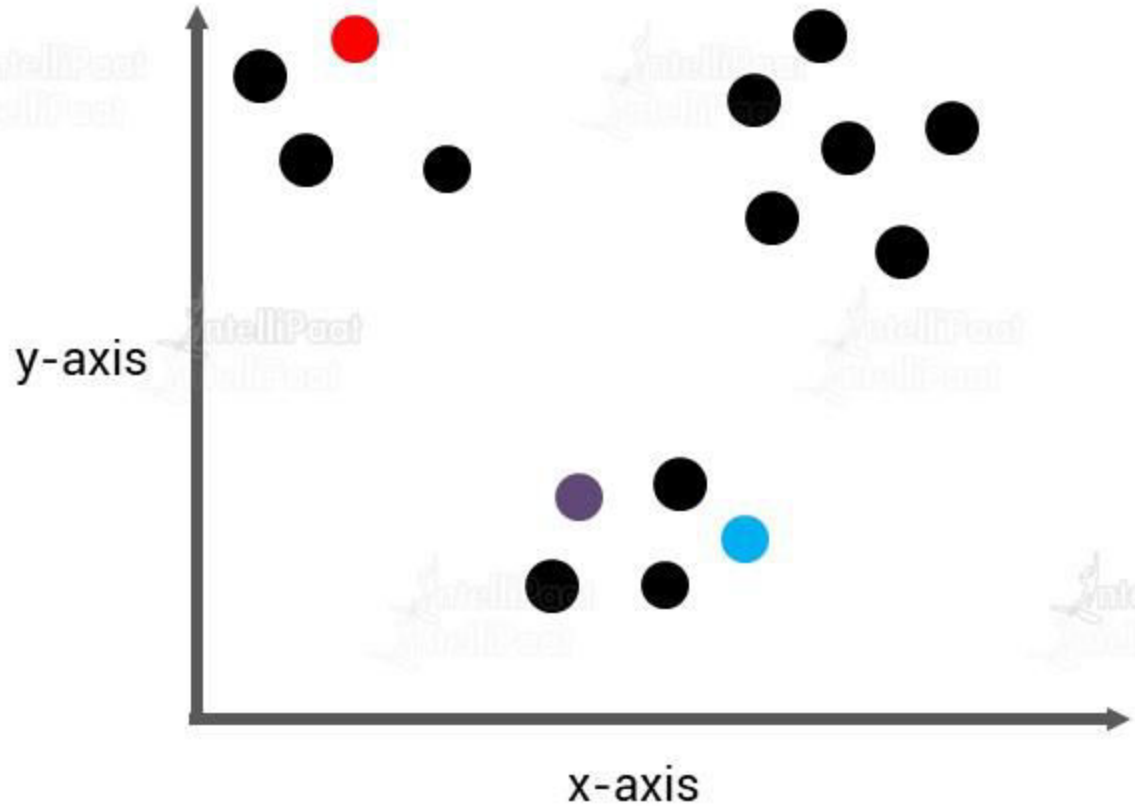
We will be using the Euclidean distance (in 2D, it is same as that of the Pythagorean theorem)



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K-means Clustering Example

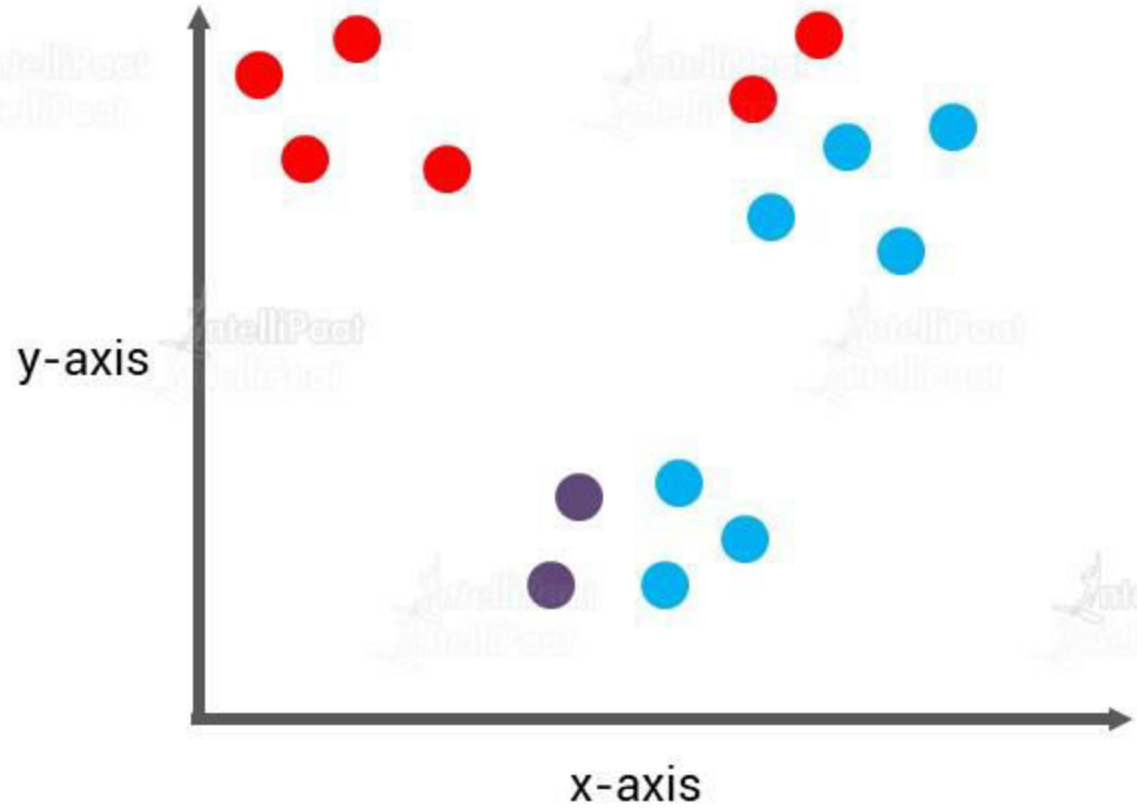
Now, we will assign the point to the nearest cluster



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K-means Clustering Example

Finally, in first iteration, we will get something like below
We have to iterate this process again and again to get the final cluster



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K-means Clustering Example

Summarizing the K-means Algorithm

```
randomly chose k examples as initial centroids
while true:
    create k clusters by assigning each
        example to closest centroid
    compute k new centroids by averaging
        examples in each cluster
    if centroids don't change:
        break
```

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Hands-on: K-means Clustering

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What is Hierarchical Clustering?

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What is Hierarchical Clustering?

This algorithm starts with all data points being assigned to a cluster of their own. Then, two nearest clusters are merged into one cluster, and this process continues. In the end, the algorithm terminates when there is only a single cluster left

Agglomerative Clustering

Each observation starts in its own cluster, and the pairs of clusters are merged into one as the algorithm moves up the hierarchy

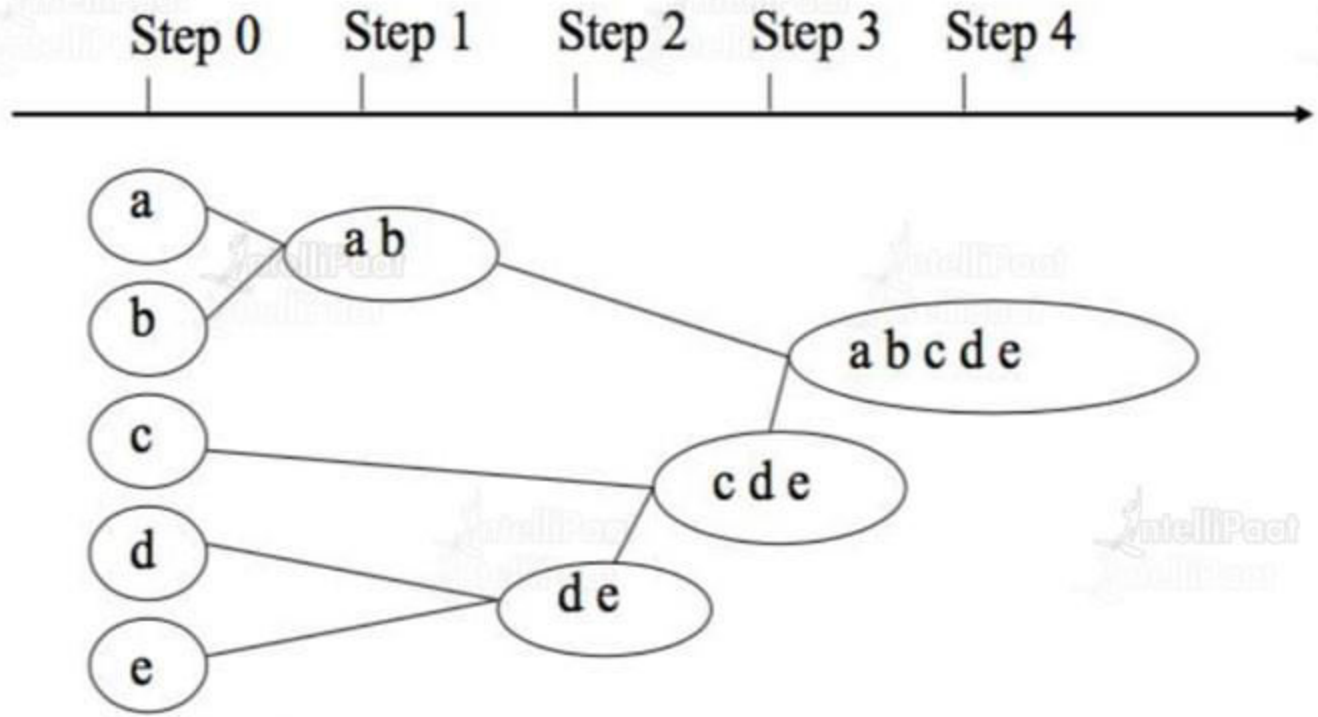
Divisive Clustering

All observations start in one cluster, and splits are performed recursively as the algorithm moves down the hierarchy

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What is Hierarchical Clustering?

Agglomerative Clustering

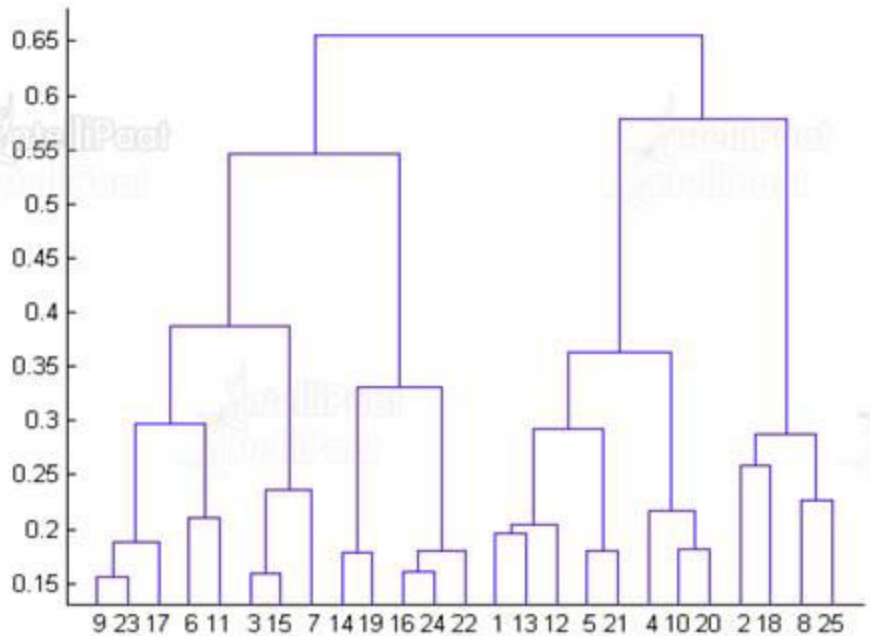


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What is Hierarchical Clustering?

Dendrogram

Graphical representation of the hierarchical clustering process



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What is Hierarchical Clustering?

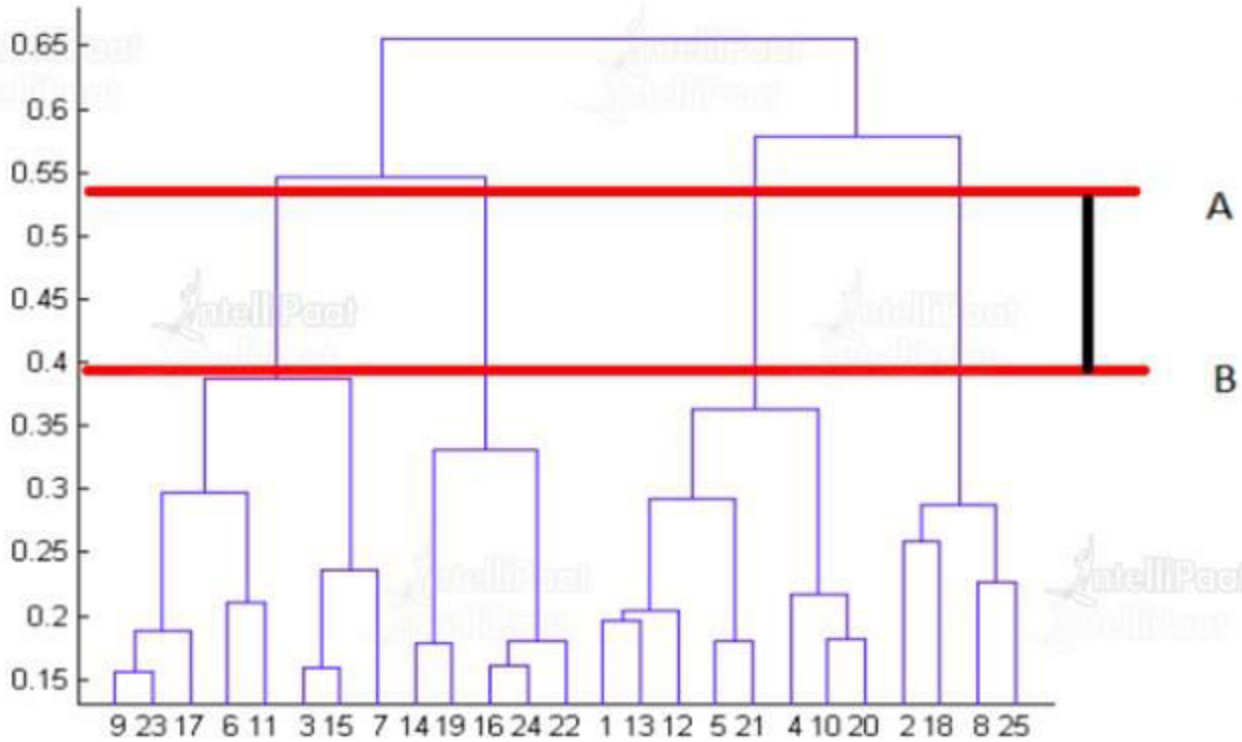
Characteristics of a Dendrogram

- The height in the dendrogram at which two clusters are merged represents the distance between two clusters in the data space
- The decision on the number of clusters that can best depict different groups can be made by observing the dendrogram
- The best choice of the number of clusters is the number of vertical lines in the dendrogram (cut by horizontal lines) that can transverse the maximum distance vertically without intersecting a cluster

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What is Hierarchical Clustering?

Dendrogram



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What is Hierarchical Clustering?

Applications of Hierarchical Clustering

- Recommendation engines
- Market segmentation
- Social network analysis
- Search result grouping
- Medical imaging
- Anomaly detection

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Hands-on: Hierarchical Clustering

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Optimal Clustering

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Optimal Clustering

Optimal clustering is used to find the optimal number of clusters (k) to be used in the clustering algorithm

Silhouette method

The silhouette value is a measure of how similar an object is to its own cluster (cohesion) compared to other clusters (separation)

Elbow method

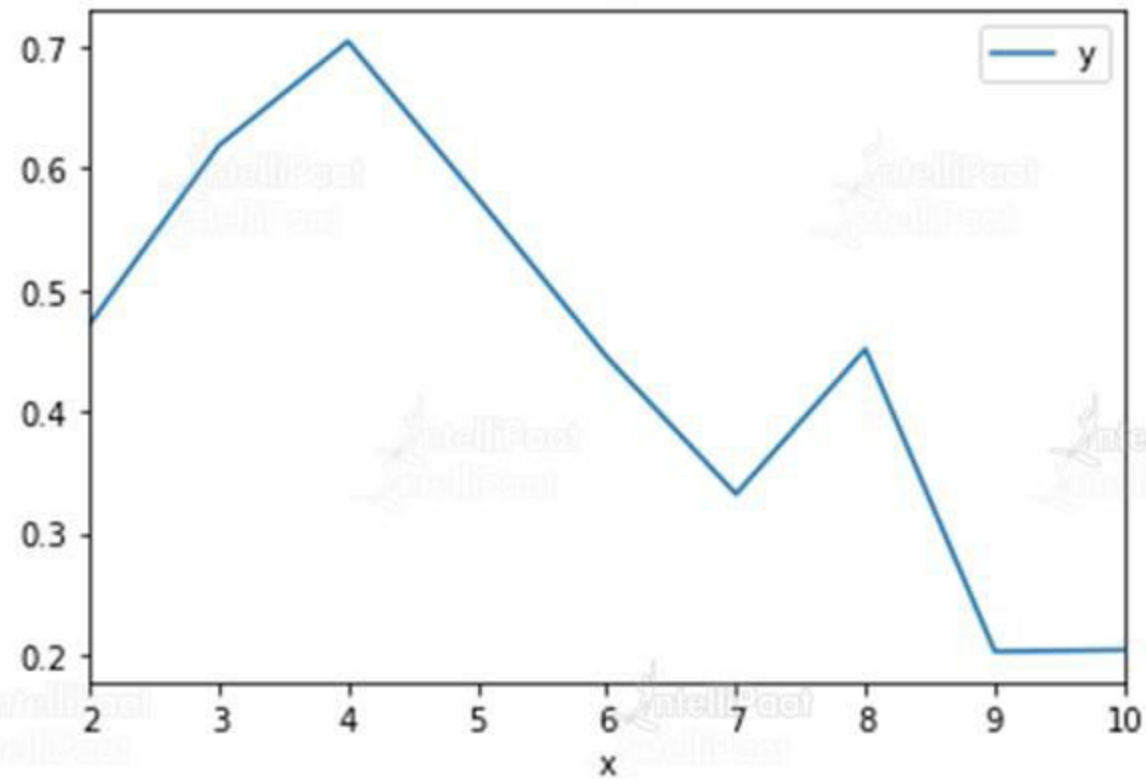
It is the cluster analysis method designed to help find the appropriate number of clusters in a dataset

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Optimal Clustering

Silhouette method

Elbow method

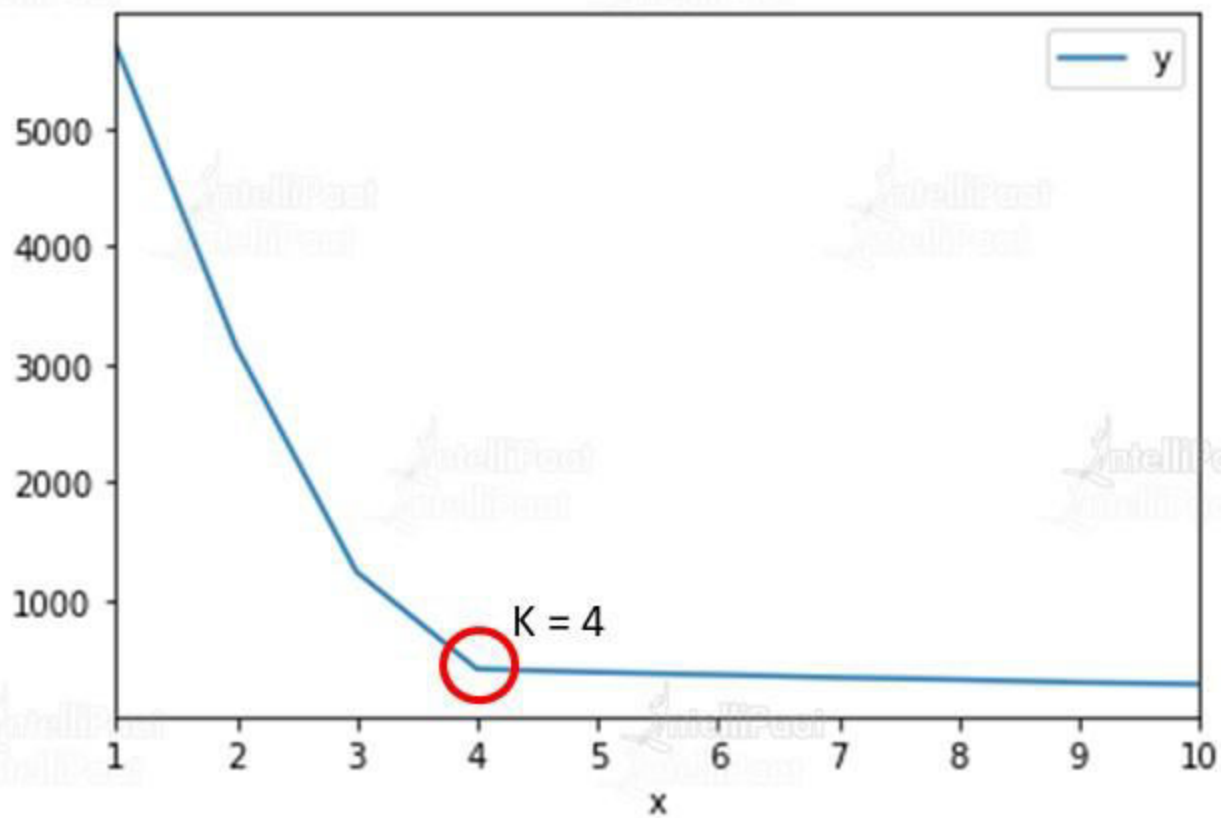


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Optimal Clustering

Silhouette method

Elbow method



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Hands-on: Optimal Clustering

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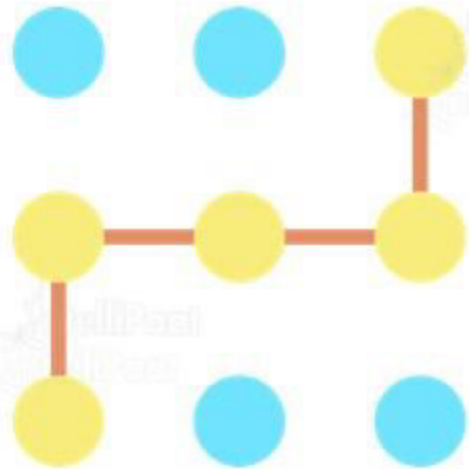
Association Rule Mining

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Association Rule Mining

A procedure aims to observe the frequently occurring patterns, correlations, or associations in datasets

In an implication, if P implies Q, then P is called the **antecedent** and Q is called the **consequent**

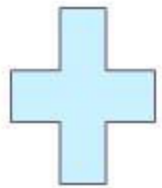


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Association Rule Mining

If a customer buys bread, then probably he/she will also buy milk







Antecedent **Consequent**

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Association Rule Mining

Market Basket Analysis

Product A + Product B + Product C might be most probably bought together





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Association Rule Mining

Market Basket Analysis



- Changing the store layout according to trends
- Customer behavior analysis
- Catalogue design
- Cross-marketing on online stores
- Sensing the trending items customers buy
- Sending customized emails with add-on sales

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Association Rule Mining

Association Rule Mining Applications



To identify value added services on mobile subscription and to know what product can be offered to a customer



To understand which symptoms tend to co-exist and to improve patient care and medicine prescription



To understand customer transition so that companies can tie up with other vendors for additional discounts



To plan efficient public services and help public businesses set up new factories, shopping malls, etc. and market particular products

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Hands-on: Association Rule Mining

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Common Ways to Measure Association

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Common Ways to Measure Association

Measure 1: Support

Measure 2: Confidence

Measure 3: Lift

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Common Ways to Measure Association

Support

Confidence

Lift

This shows how popular an itemset is by checking the proportion in the dataset

$$\text{Support}\ (x \Rightarrow y\ ) = \frac{N_{x \wedge y}}{N_{x U y}}$$

- *N* is the number of instances
- *Nx* is the number of instances covering *X*
- $Nx \wedge y = NxUy$ is the number of instances covering both *X* and *Y*

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Common Ways to Measure Association

- Support
- Confidence
- Lift

Cappuccino	Latte	Espresso	Americano	Ristretto	Tea	Muffin	bagel
1	0	0	0	0	0	1	0
0	2	0	0	0	0	1	1
0	0	1	0	0	0	0	0
1	0	0	0	0	0	0	0
0	0	0	0	0	1	2	0
0	0	0	1	1	0	0	0
...	...	...	...	...	...	...	...

$$\{\text{Tea, Latte}\} \Rightarrow \{\text{muffin}\}$$

Frequency does not matter, only presence matters

$$\text{Support}\ (\{\text{Tea, Latte}\} \Rightarrow \{Muffin\}) = \frac{N_{\{Tea,\ Latte,\ Muffin\}}}{N}$$

Value between 0 = Bad and 1 = Good

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Common Ways to Measure Association

Support

Confidence

Lift

This shows how likely an item (Item A) will be bought when another item (Item B) is purchased

$$\text{Confidence}\ (x \Rightarrow y) = \frac{N_{x \wedge y}}{N_x} = \frac{N_{xUy}}{N_x}$$

$$\text{Confidence}\ (\{\text{Tea, Latte}\} \Rightarrow \{Muffin\}) = \frac{N_{\{Tea,\ Latte,\ Muffin\}}}{N_{\{tea,Latte\}}}$$

Value between 0 = Bad and 1 = Good

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Common Ways to Measure Association

Support

Confidence

Lift

This shows the importance of an item w.r.t. its lift value. Higher lift, higher importance

$$\text{Lift}\ (x \Rightarrow y) = \frac{N x \wedge y \;\; N}{(N_x \;\; N)(N_y \, N)} = \frac{N_{x \wedge y} \, N}{N_x \, N_y}$$

$$\text{Lift}\ (\{\text{Tea, Latte}\} \Rightarrow \{Muffin\}) = \frac{N_{\{Tea,\ Latte,\ Muffin\}} N}{N_{\{tea,Latte\}} \, N_{\{Muffin\}}}$$

Lift $(x \Rightarrow y) > 1$ if X and Y are positivly correlated
Lift $(x \Rightarrow y) \approx 1$ if X and Y are independent
Lift $(x \Rightarrow y) < 1$ if X and Y are negitivly correlated

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Common Ways to Measure Association

Support

Confidence

Lift

1. Serves as quantity criteria:
   i. Can be used to filter rules Apriori
   ii. Can be used to sort rules
2. Important because there may be many rules
3. Typically, one is interested in rules:
   i. That have **support as high as possible** (the value depends on the number of items)
   ii. That have **confidence close to 1**
   iii. That have **lift higher than 1** (positive correlation)

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Common Ways to Measure Association

Support

Confidence

Lift

Suppose, 100 customers buy diapers and/or beer:

- 9 customers buy just Hoegaarden
- 40 customers buy just Pampers
- 50 customer buy just Pampers and Dommelsch
- 1 customer buys Pampers, Hoegaarden, and Dommelsch

Consider the following four rules:

{ Pampers } ⇒ { Dommelsch }

{ Pampers } ⇒ { Hoegaarden }

{ Dommelsch } ⇒ { Pampers }

{ Pampers, Dommelsch} ⇒ { Hoegaarden }

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Common Ways to Measure Association

Support

Confidence

Lift

First rules:

{ Pampers } ⇒ { Dommelsch }

$$\text{Support}(\{\text{Pampers}\} \Rightarrow \{\text{Dommelsch}\}) = \frac{N_{\{Pampers,Dommelsch\}}}{N} = \frac{51}{100} = 0.51$$

$$\text{Confidence}(\{\text{Pampers}\} \Rightarrow \{\text{Dommelsch}\}) = \frac{N_{\{Pampers,Dommelsch\}}}{N_{\{Pampers\}} N_{\{Dommelsch\}}} = \frac{51}{91} = 0.51$$

$$\text{Confidence}(\{\text{Pampers}\} \Rightarrow \{\text{Dommelsch}\}) = \frac{N_{\{Pampers,Dommelsch\}} N}{N_{\{Pampers\}}} = \frac{51*100}{91*51} = 1.10$$

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Common Ways to Measure Association

Support

Confidence

Lift

{ Pampers, Dommelsch} ⇒ { Hoegaarden }

Answer	N_X	N_Y	N_{XY}	support	confidence	lift
$\{Pampers\} \Rightarrow \{Dommelsch\}$	91	51	51	51/100 = 0.51	51/91 = 0.56	(51*100)/(91*51) = 1.10
$\{Pampers\} \Rightarrow \{Hoegaarden\}$	91	10	1	1/100 = 0.01	1/91 = 0.01	(1*100)/(91*10) = 0.11
$\{Dommelsch\} \Rightarrow \{Pampers\}$	51	91	51	51/100 = 0.51	51/51 = 1	(51*100)/(51*91) = 1.10
$\{Pampers, Dommelsch\} \Rightarrow \{Hoegaarden\}$	51	10	1	1/100 = 0.01	1/51 = 0.02	(1*100)/(51*10) = 0.20

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